

Weeks 2-10

Extra / grades 6-7

Optional investigations; offer the next page by readiness.

Week 2 is one shared collection; grade bands below apply to Weeks 3-10.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Each student set includes the same Week 2 library. Print its chosen pages once.

Week and investigation	PDF pages
02 Switches and lamps	2-43
03 Repeating secret machines	44-46
04 Around the ring	47-49
05 Tower cities	50-51
06 Code-breaking with guarantees	52-54
07 Take-away games	55-57
08 Rooks and piles	58-60
09 Bouncing paths	61-63
10 Bridges and delivery routes	64-66

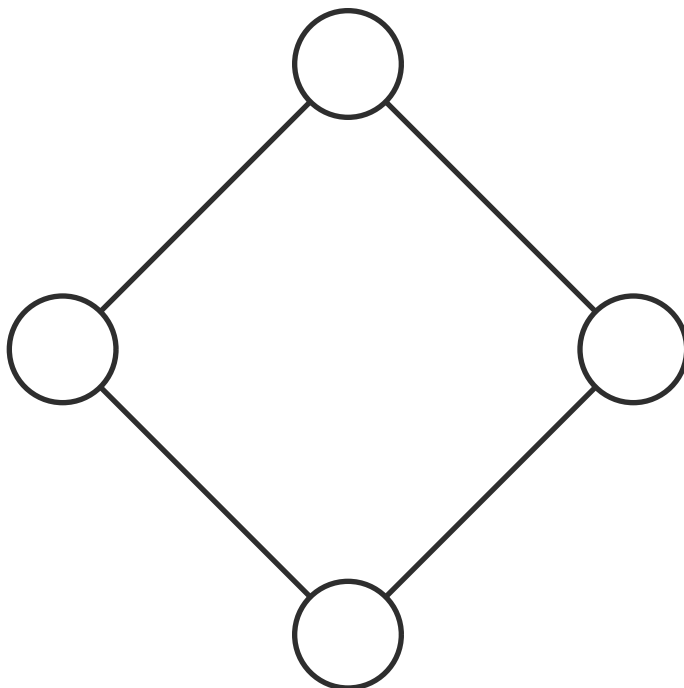
Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

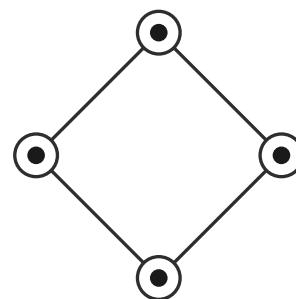
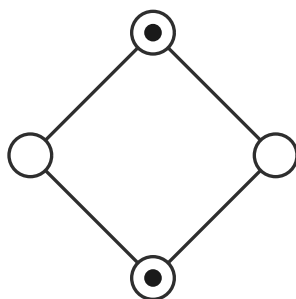
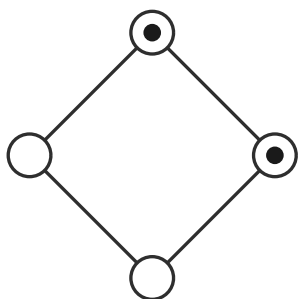
Weekly footers retain their own page numbers and activity IDs for the use log.

Unpiloted. Week 2 updated September 28; other core revisions September 27.

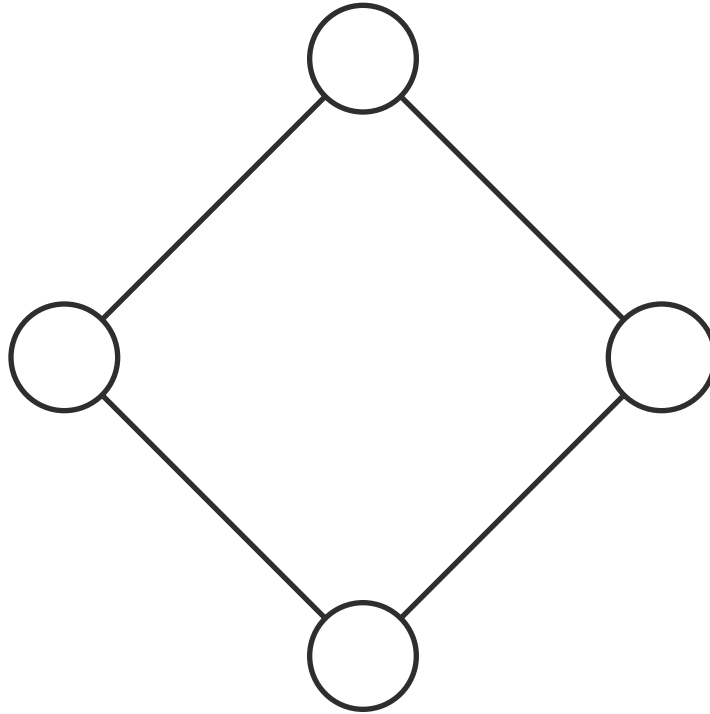
Problem 1: Empty is OFF. A dot is ON. Choose a line. Change BOTH lamps at its ends: add a dot or erase a dot. Try a move. Try that same move again.



Problem 2: Start all OFF each time. Make each small picture on the big board.

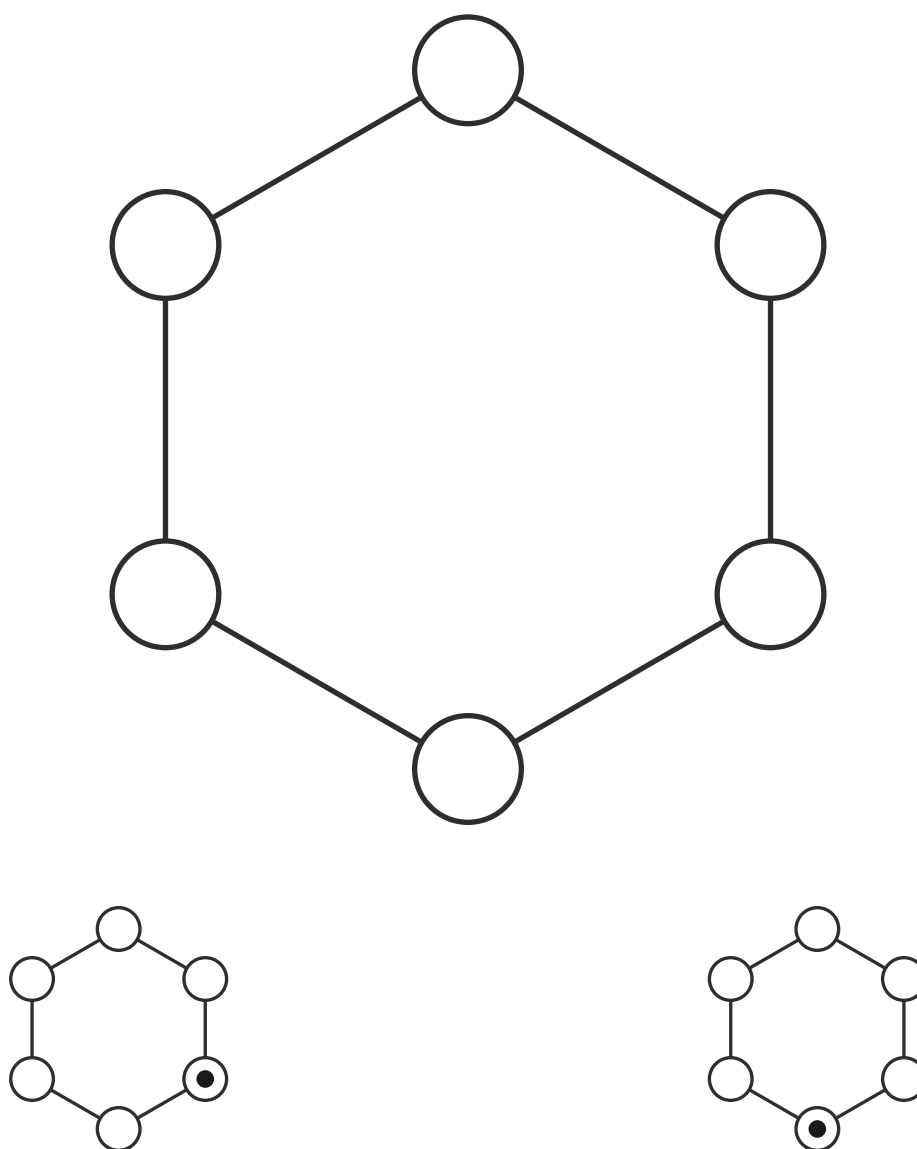


Problem 3: Play with a partner. Start all OFF. Make one move: change both ends of a line. Can your partner turn all the lamps OFF? Swap jobs.



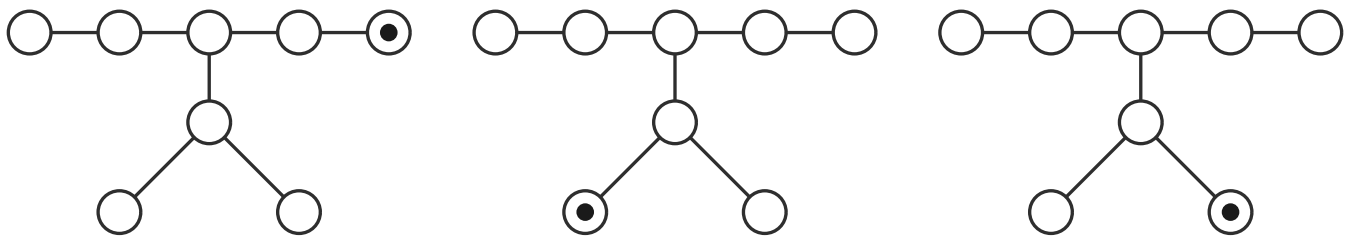
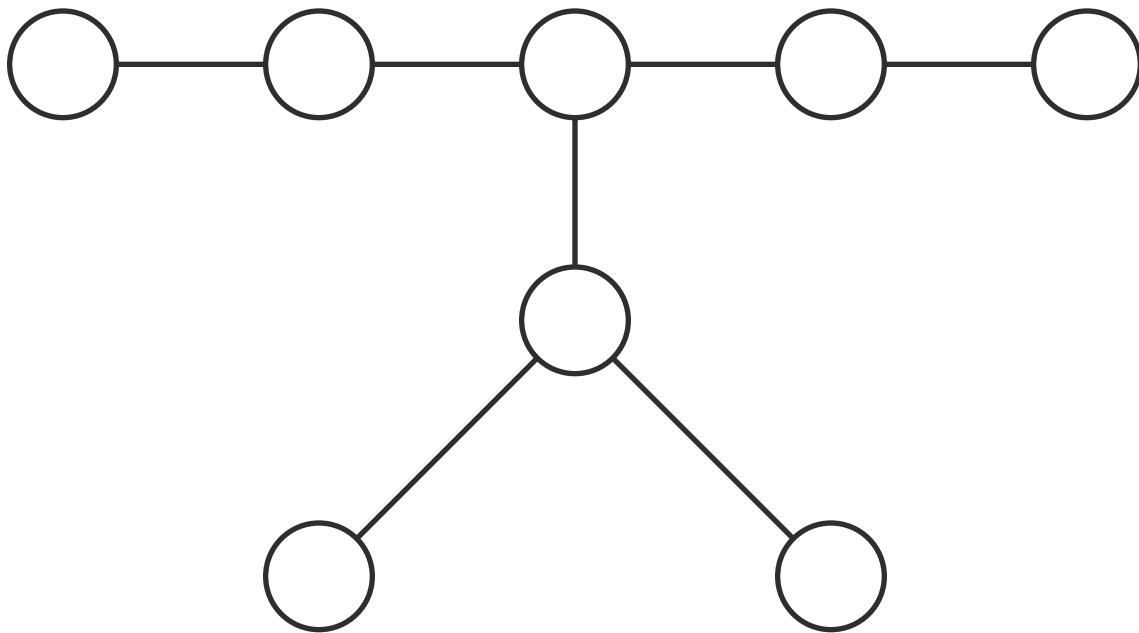
Problem 4: Start all OFF. Make TWO moves for your partner to undo. Swap jobs. Then try THREE moves. Show your moves if your partner wants a hint.

Problem 5: Start with only the top lamp ON each time. Change both ends of a line to make each small picture. Keep exactly ONE lamp ON after every move.

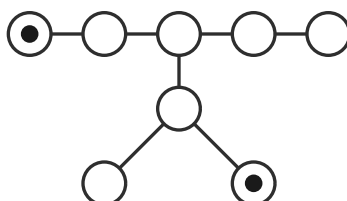


Problem 6: Start at the top again. Visit every lamp and bring the light home.

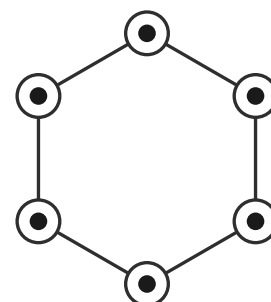
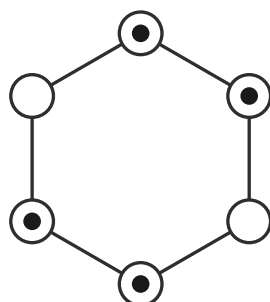
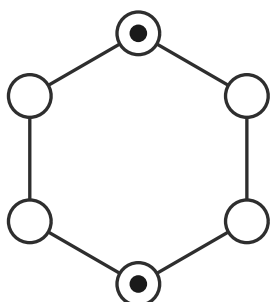
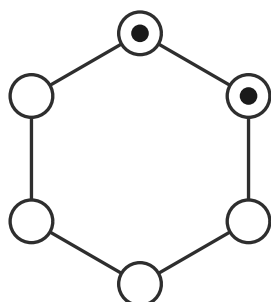
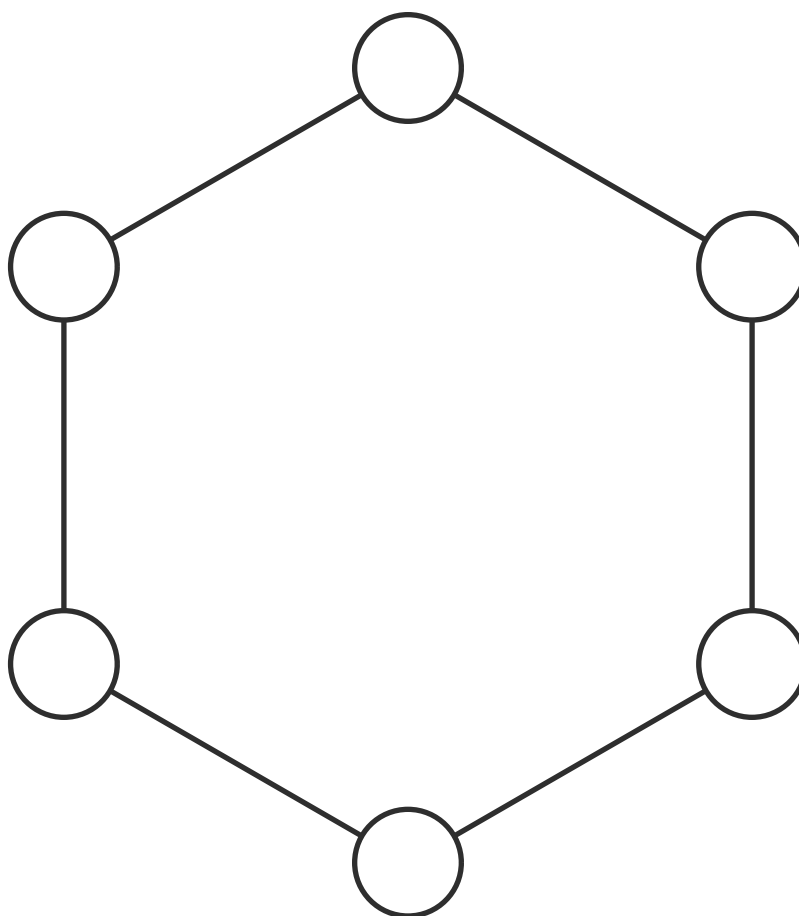
Problem 7: Start with only the far-left lamp ON each time. Change both ends of a line. Move the light to match each small picture. Keep ONE lamp ON.



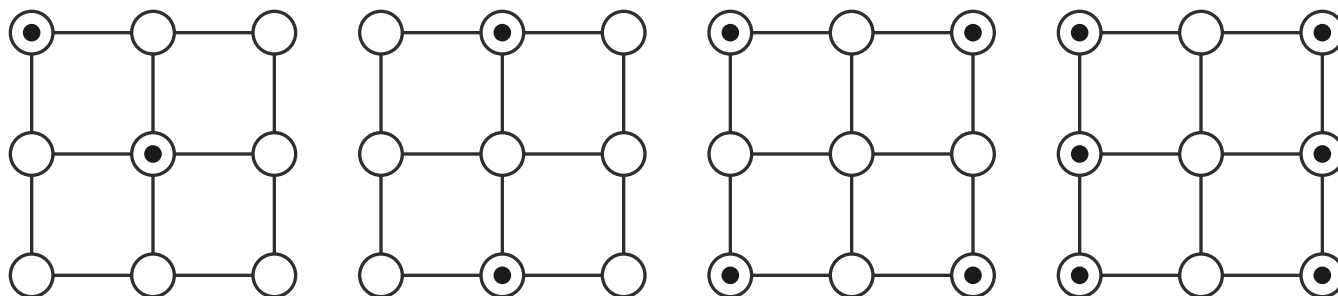
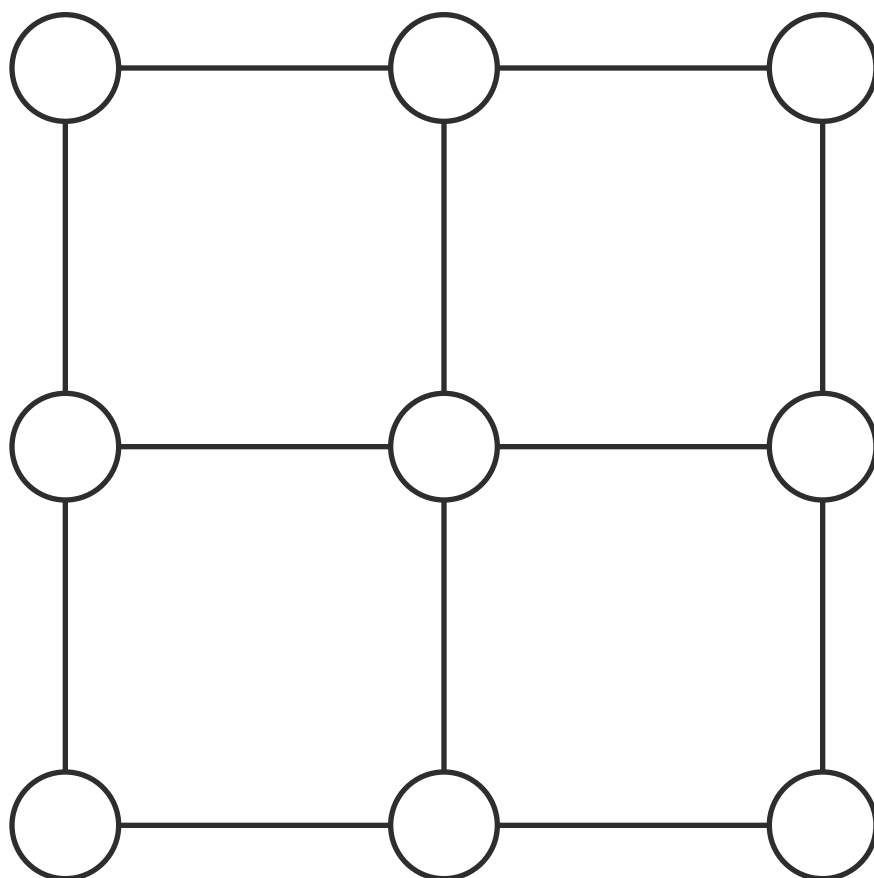
Problem 8: Now start all OFF. Make this two-light picture. You may have more than one light ON.



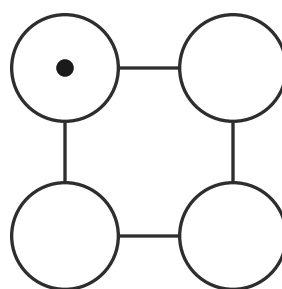
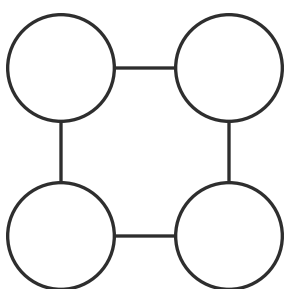
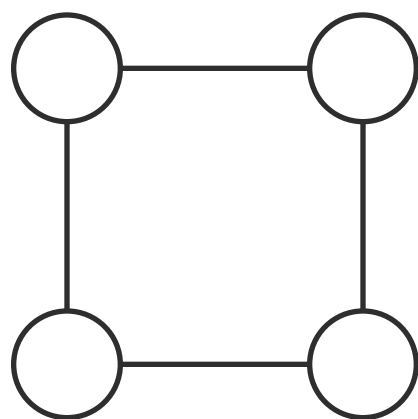
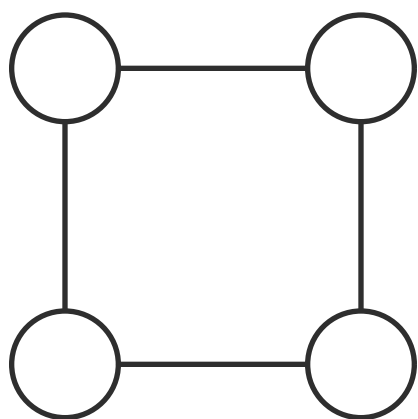
Problem 9: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Point to one you want to do again.



Problem 10: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Try a different way to make one picture.



Problem 11: Put a dot in the top-left lamp of the LEFT island. Change BOTH ends of one line each move. Can you make the small picture?



Problem 12: Draw ONE new line between lamps on different islands. Start again with only the top-left lamp of the left island ON. Try the trip with your bridge.

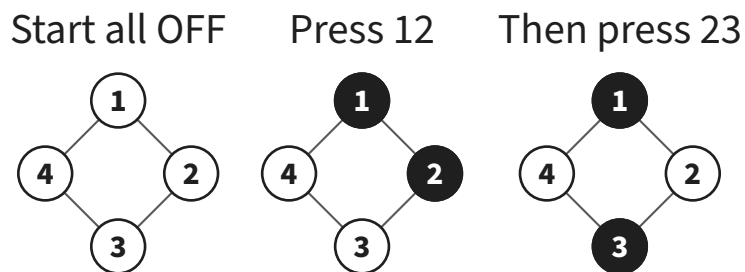
Problem 13: Draw 4 to 8 lamps. Join them with lines so every lamp has a path to every other lamp. Keep lines from crossing. Start all OFF. Make TWO moves, changing both ends of a line each time. Let a partner turn them all OFF.



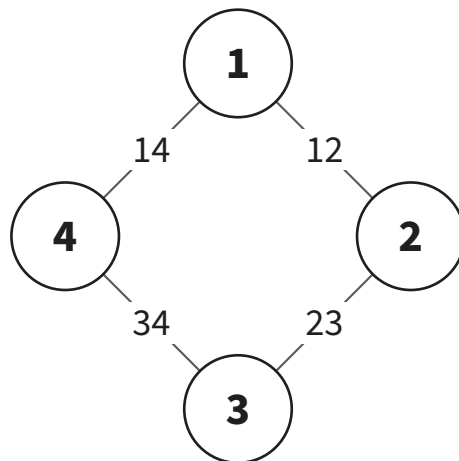
Problem 14: Add one new lamp. Join it to an old lamp with a line. Make a new puzzle for your partner.

Problem 15: Use the line names to save your moves. Start all OFF on the big board. Press a line: change *both* end lamps. Add or erase pencil dots, or flip two-sided counters. Dotted or dark means ON.

The line joining lamps 1 and 2 is called **12**. Replay these two moves on the big board.



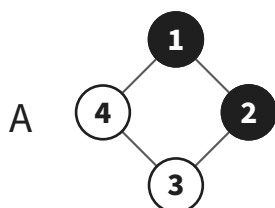
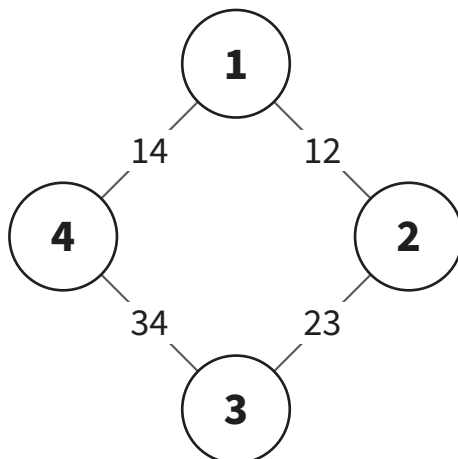
Save the moves in order: **12, 23**. You may ask someone to write while you point.



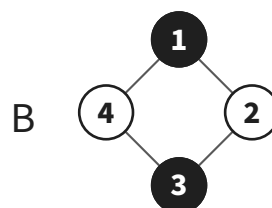
Get back to all OFF by pressing lines. Save your moves so a partner can replay them.

Moves: _____

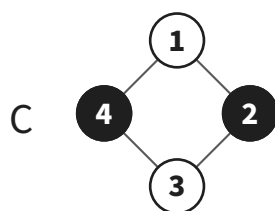
Problem 16: Make each target picture on the big board. Reset all lamps to OFF before each target. Change both ends of one line at a time. Write your moves in order.



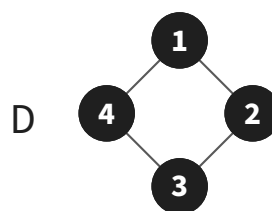
Moves: _____



Moves: _____



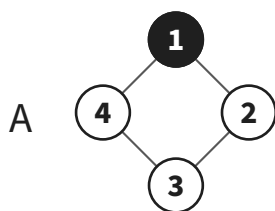
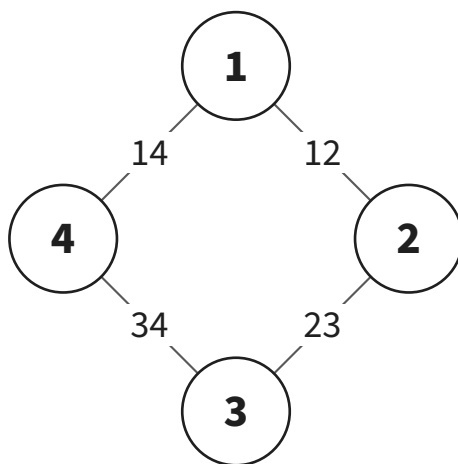
Moves: _____



Moves: _____

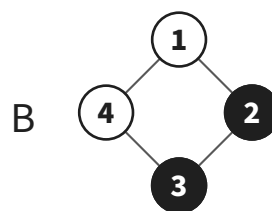
Give a partner one of your move lists. Reset all OFF. Have your partner replay it and point to its target.

Problem 17: Try each target picture. Start all OFF each time. Change both ends of one line at a time. Write a move list that works, or check “not yet.”



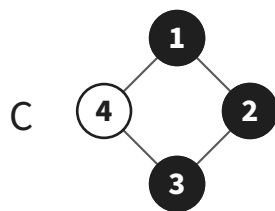
Moves: _____

☐ Not yet



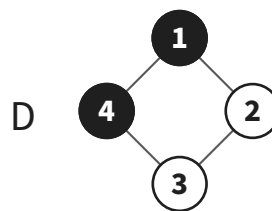
Moves: _____

☐ Not yet



Moves: _____

☐ Not yet

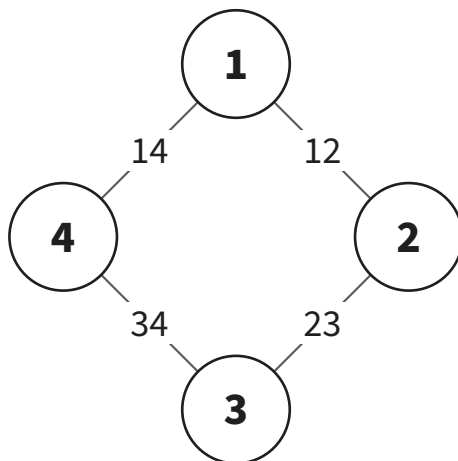


Moves: _____

☐ Not yet

For a “not yet” picture, try a different move list on the board. Keep any list that works.

Problem 18: For each row, set up its starting picture by hand on the big board: add dots or turn counters over. Leave lamps 3 and 4 OFF. Then press 12 once. Darken the ON lamps in the answer picture. Count ON lamps before and after.



A  — 12 — 
ON before: _____

press 12 →

 — 12 — 
ON after: _____

B  — 12 — 
ON before: _____

press 12 →

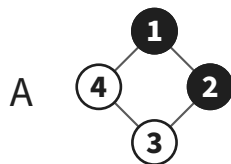
 — 12 — 
ON after: _____

C  — 12 — 
ON before: _____

press 12 →

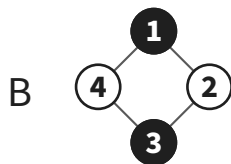
 — 12 — 
ON after: _____

Problem 19: These are the targets you tried. For each picture, circle “made” or “not yet.” Count its ON lamps. Use the big board to try any picture again; reset all OFF each time.



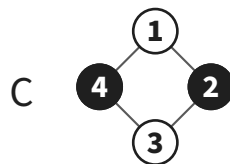
ON: _____

made / not yet



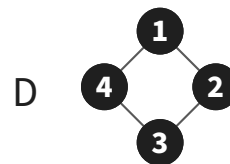
ON: _____

made / not yet



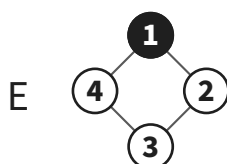
ON: _____

made / not yet



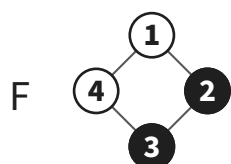
ON: _____

made / not yet



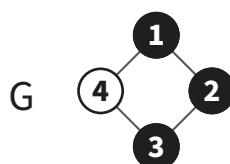
ON: _____

made / not yet



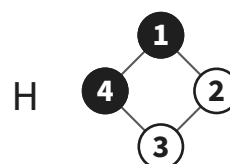
ON: _____

made / not yet



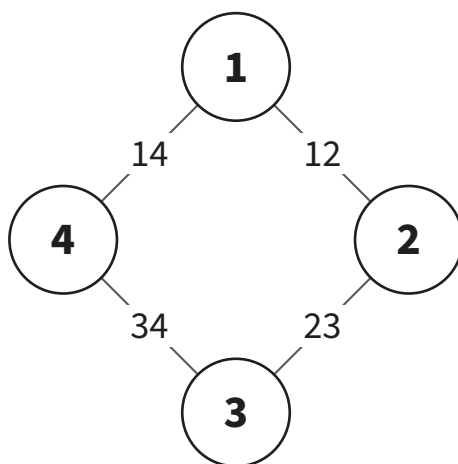
ON: _____

made / not yet



ON: _____

made / not yet

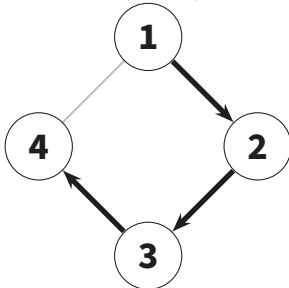


Compare the ON counts in the pictures you made.

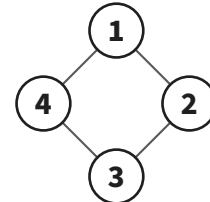
Point to what is alike. Then compare the “not yet” pictures.

Problem 20: Follow each arrow path on the big board: change both ends of each line, in order. Start all OFF for each path. Darken the ON lamps in its final picture.

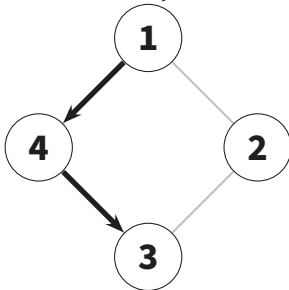
A: 12, 23, 34



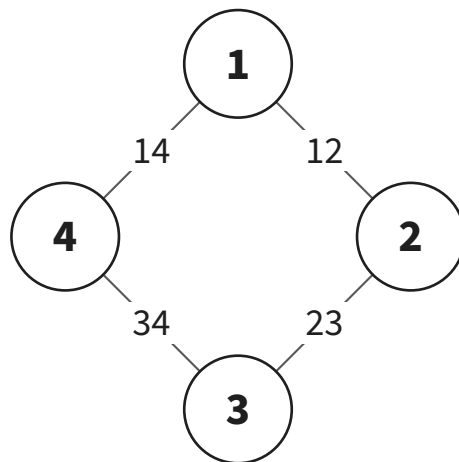
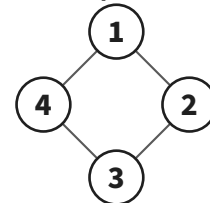
Final picture:



B: 14, 43

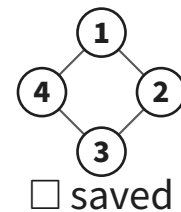
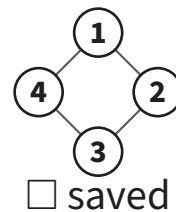
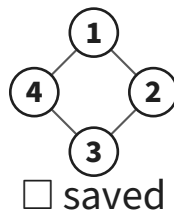
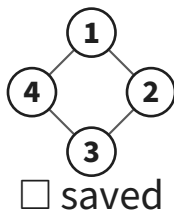
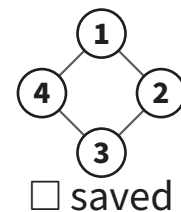
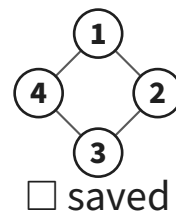
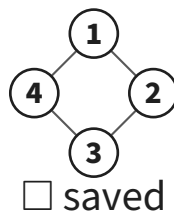
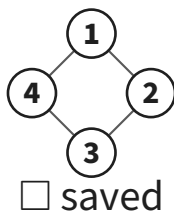
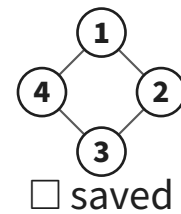
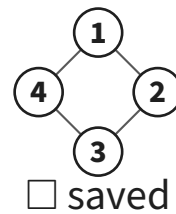
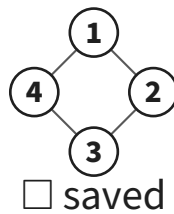
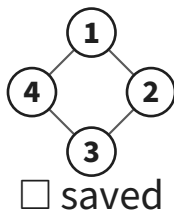
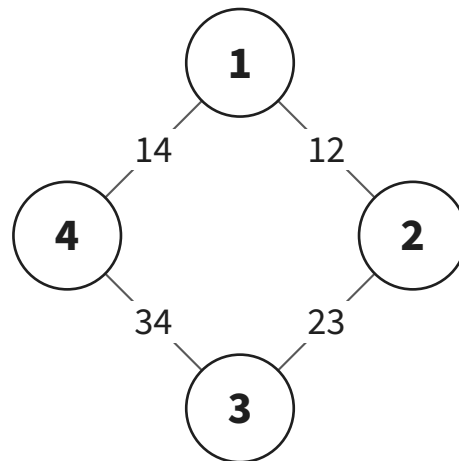


Final picture:



Replay one path slowly. Point to each lamp that changes twice. Point to each lamp changed just once.

Problem 21: Start all OFF on the big board. Try some moves. Copy the result onto a little ring: darken its ON lamps. Reset and make different pictures. Include all OFF. Leave extra rings empty.



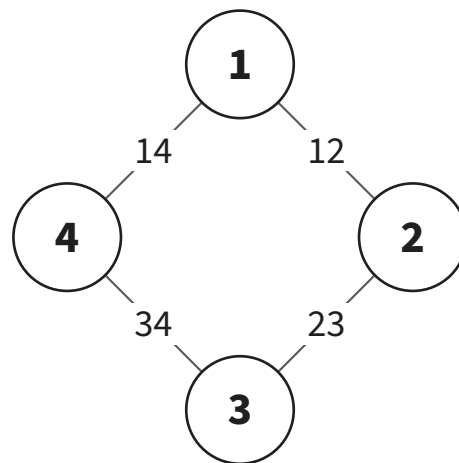
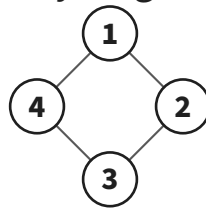
Check “saved” under each picture you record, including all OFF. This keeps unused rings separate.

Problem 22: Look at the pictures you saved in Problem 21. Count the ON lamps in each. Write that count beside its little ring. Point to the pictures with the same count. Count how many different pictures you have.

I saved _____ different pictures.

Choose a picture whose moves you have not written down. Copy it here. Start all OFF on the big board and find a move list that makes it.

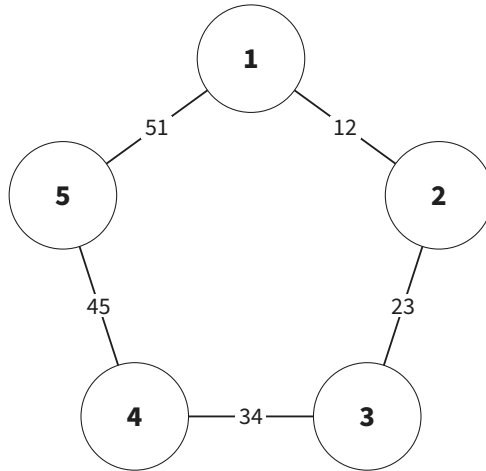
My target:



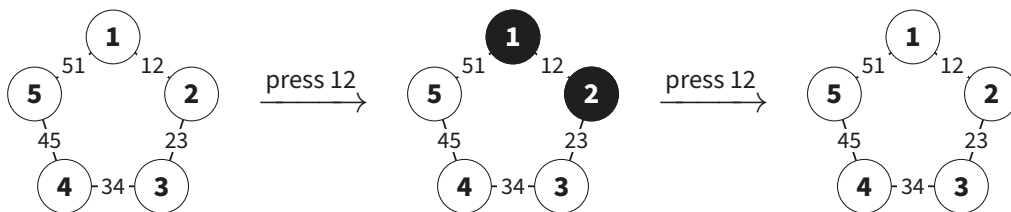
My moves: _____

Reset all OFF and ask a partner to replay your list. Compare the result with your target.

Problem 23: Try a five-lamp ring. Start all OFF on the large board. Press a line: change both end lamps. Add or erase pencil dots, or flip two-sided counters. OFF becomes ON; ON becomes OFF.



Write **12** for the line joining lamps 1 and 2. Dotted or dark means ON; white means OFF. You may ask someone to write your moves while you point.



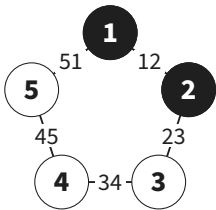
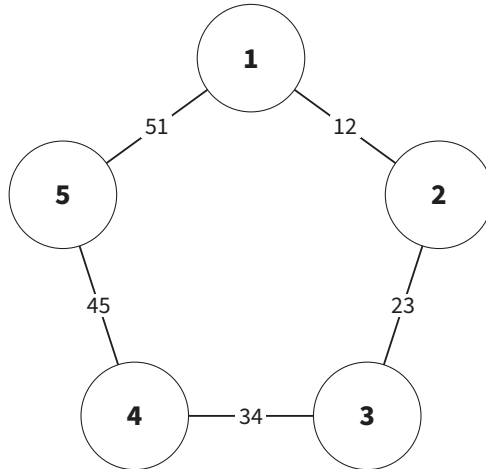
Try the two presses shown above. Then start all OFF and choose two presses of your own. Write them in the order you press them.

First press: _____

Second press: _____

Get back to all OFF. Write the presses you used.

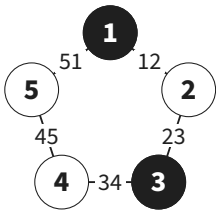
Problem 24: Start all OFF for each target below. Make the dark lamps ON and the white lamps OFF. Write two different move lists for each target. Circle the shorter list, or write “same” if they tie.



Make this picture.

First list: _____

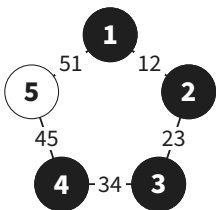
Another list: _____



Make this picture.

First list: _____

Another list: _____



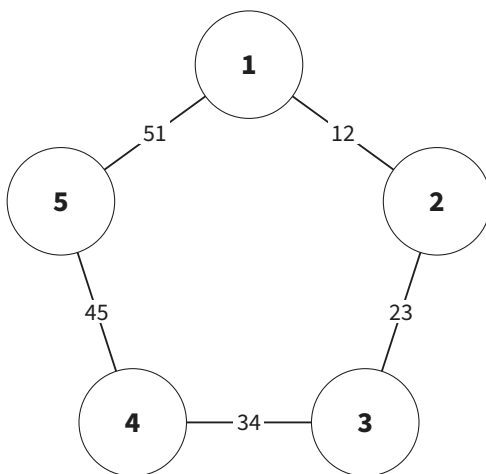
Make this picture.

First list: _____

Another list: _____

Choose one target. Test single presses, resetting to all OFF before each press. Can one press make that target? Tell a partner what happened.

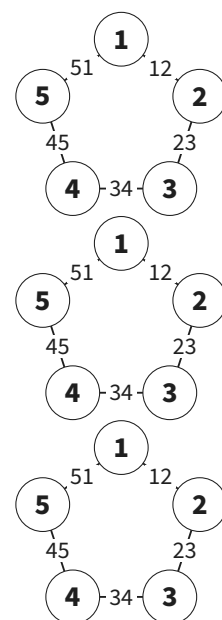
Problem 25: Start all OFF for each move list. Read from left to right. Press one line at a time, then color the lamps that finish ON in its small ring.



Press **12**, then **23**.

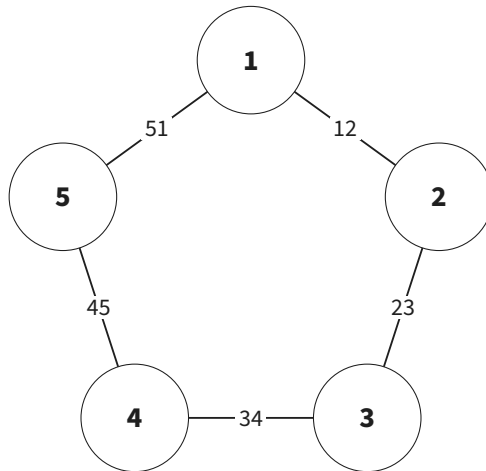
Press **23**, then **12**.

Press **12**, then **23**, then **12**, then **12**.



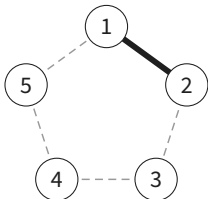
Compare the three pictures. In the last list, cross out two copies of the same press. Reset and try the presses left over. Do you get the same picture?

Problem 26: For each pair, start all OFF. Press each thick line on the left once, then color the lamps that finish ON. Reset to all OFF. Test the thick lines on the right and color that final picture.



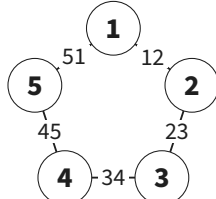
The dashed lines are the lines you do not press. In the first pair, try line 12 on the left; on the right try 23, 34, 45, 51.

Press these

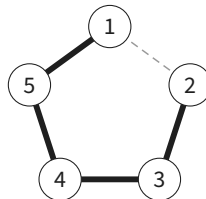


___ presses

Color result

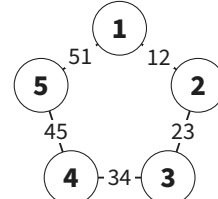


Reset. Press these

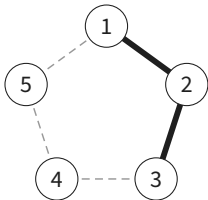


___ presses

Color result

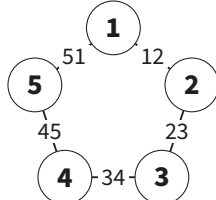


Press these

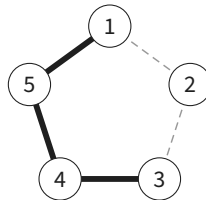


___ presses

Color result

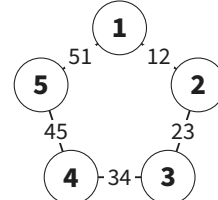


Reset. Press these



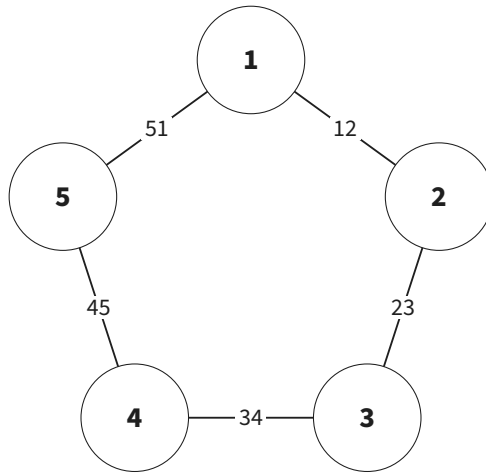
___ presses

Color result

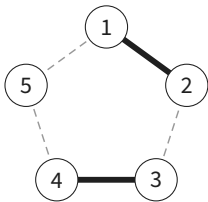


In each pair, count the presses and compare the two final pictures. Circle the side with fewer presses.

Problem 26 (continued): Try two more pairs. Begin all OFF each time. Press each thick line once and color the lamps that finish ON. Count the presses.

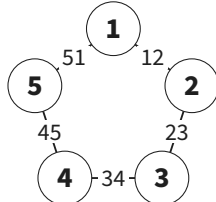


Press these

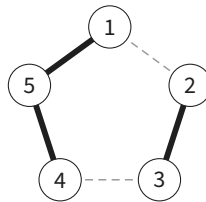


___ presses

Color result

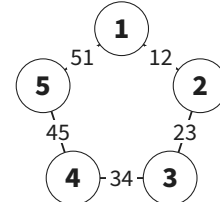


Reset. Press these

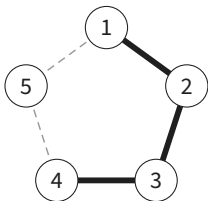


___ presses

Color result

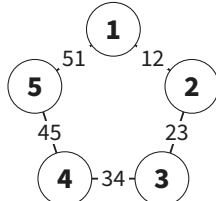


Press these

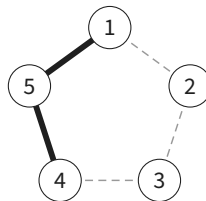


___ presses

Color result

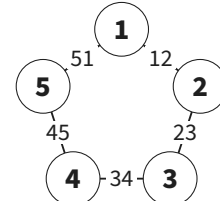


Reset. Press these



___ presses

Color result

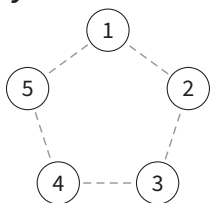


Compare the two final pictures in each pair. Circle the side with fewer presses.

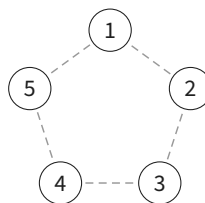
Finish this sentence with words or a drawing: “When I press the other lines, ...”

Problem 27: Make your own pair of line pictures. Choose one, two, or three lines on the left ring and trace them thickly. On the right ring, trace every line you did not choose.

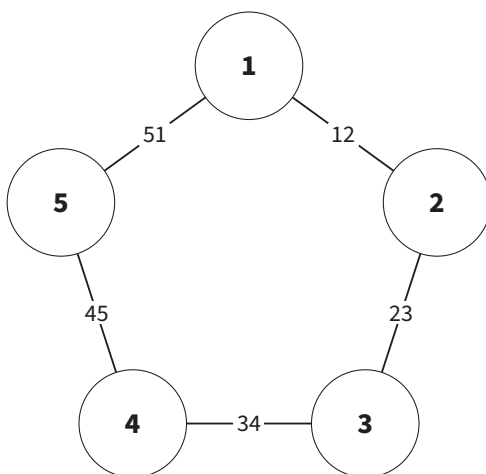
My chosen lines



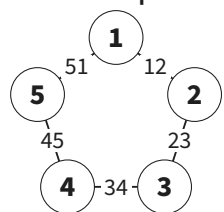
The other lines



Start all OFF on the large mat. Press each chosen line once. Color the first final picture below. Reset and test the other lines; color the second final picture.

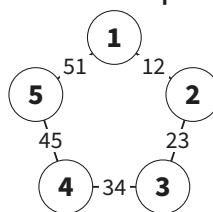


First final picture



_____ presses

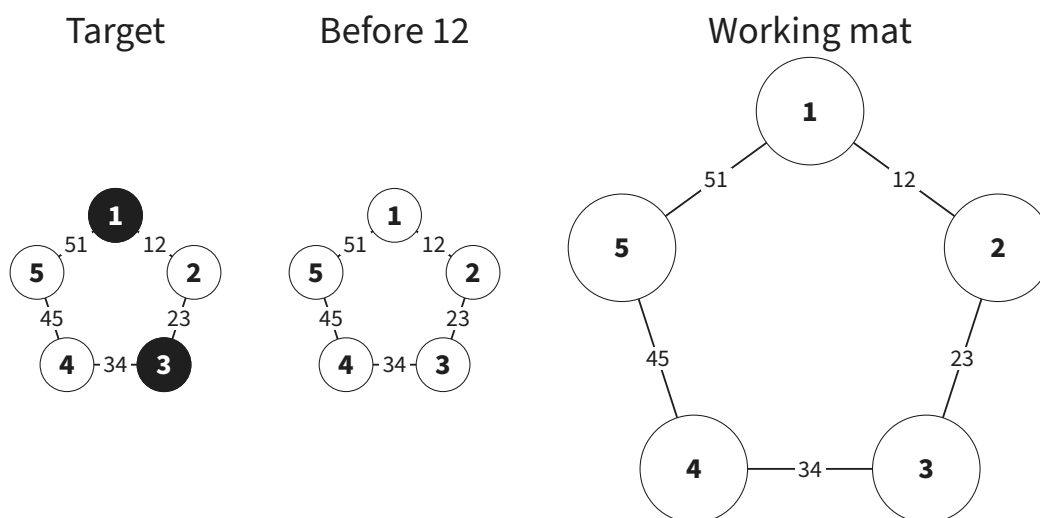
Second final picture



_____ presses

Compare your pictures. Show your partner the pair and ask them to test it.

Problem 28: Make the target below by fixing one lamp at a time. Start all OFF. This time, do not press line 51. Use each line at most once.



Lamp 1 is OFF, but the target needs it ON. **Press 12 now.** Circle “yes” in the first row. Lamp 1 now matches the target.

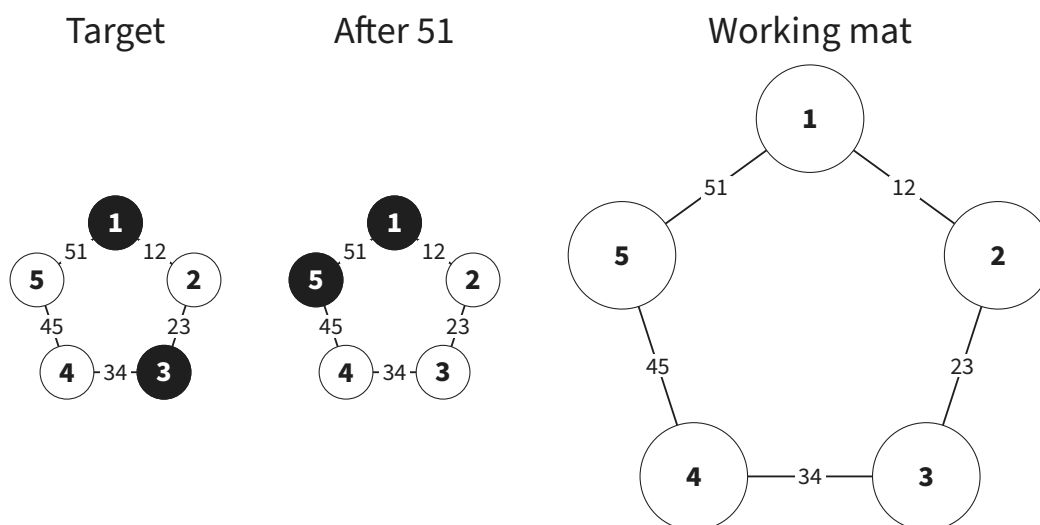
Next look at lamp 2. If it differs from the target, press 23; if it matches, leave 23 alone. Do the same for lamps 3 and 4, using the next line each time.

Look at lamp	To match the target, press this line?	Circle one	
1	12	yes	no
2	23	yes	no
3	34	yes	no
4	45	yes	no

Stop after 45. Does lamp 5 match the target? ☐ yes ☐ no

Write the lines you pressed, in order.

Problem 28 (continued): Try the same target again. Start all OFF. This time, press line 51 first. Then decide about 12, 23, 34, 45, in that order. Use each line at most once.



After pressing 51, lamp 1 is already ON, just as the target needs. **Do not press 12.** Circle “no” in the first row.

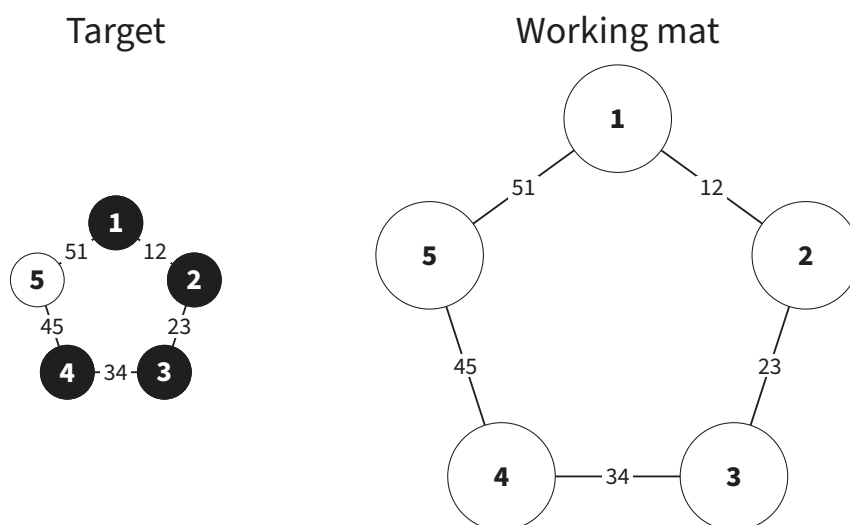
For each next lamp, compare it to the target. Press the next line only if that lamp needs to change.

Look at lamp	To match the target, press this line?	Circle one	
1	12	yes	no
2	23	yes	no
3	34	yes	no
4	45	yes	no

Stop after 45. Does lamp 5 match the target? ☐ yes ☐ no

Write your whole move list, starting with 51.

Problem 29: Make this new target in two ways. Start all OFF for each trial. Decide about 51 first, then 12, 23, 34, 45. Press each line at most once.



Look at each lamp in order. If it differs from the target, press the line below it. If it matches, do not press that line. After lamp 4, stop and check lamp 5. Include 51 in your press count if you used it.

First trial: leave 51 alone.

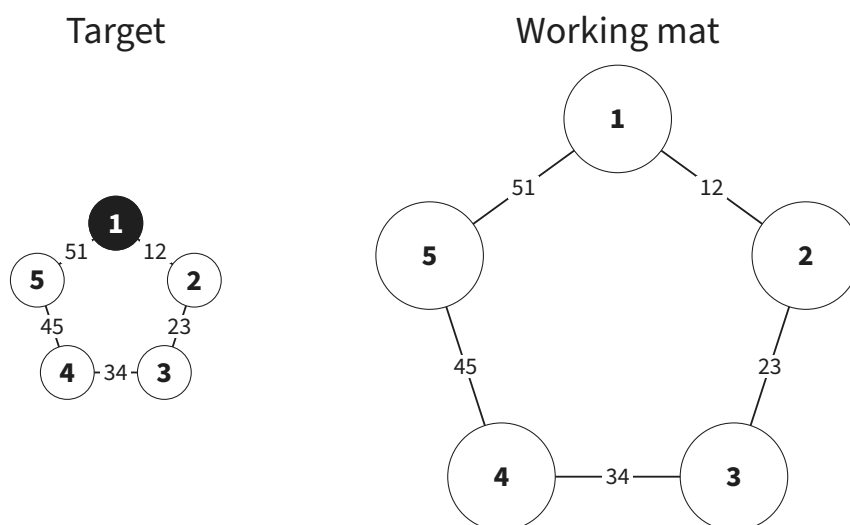
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Reset to all OFF. Second trial: press 51 first.

Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Circle the trial that uses fewer presses. Point to its lines on the working mat.

Problem 30: Try this target: only lamp 1 ON. Start all OFF for each trial. Choose about 51 first, then decide about 12, 23, 34, 45. Press each line at most once.



At each lamp, press the next line only if that lamp needs to change to match the target. After deciding about 45, stop. Check lamp 5 carefully. Include 51 in your press count if you used it.

First trial: leave 51 alone.

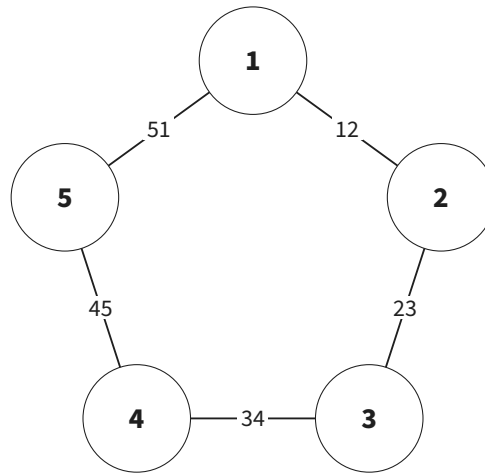
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Reset to all OFF. Second trial: press 51 first.

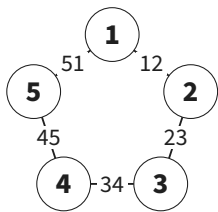
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Put a cross beside a trial if its final picture does not match the target. Show a partner which lamp does not match.

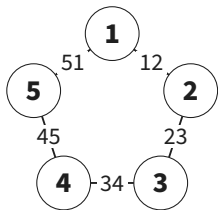
Problem 31: Invent three new targets. In each small ring, color two or four lamps ON. Test each target on the large mat, starting all OFF each time.



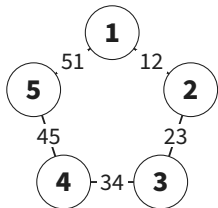
Find one move list using each line at most once. For the other list, use every line missing from the first list, once each. Test both. Count the presses and circle the shorter list.



My target. Counts: _____
 First list: _____
 Another list: _____



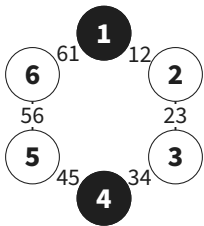
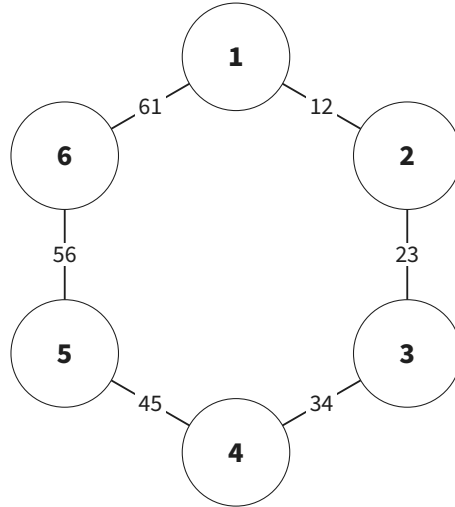
My target. Counts: _____
 First list: _____
 Another list: _____



My target. Counts: _____
 First list: _____
 Another list: _____

If you want another puzzle: can you draw a target with two or four ON lamps that needs more than two presses?

Problem 32: Start all OFF on this six-lamp ring. Follow one route between the dark lamps of a target. Press each line on that route once. Reset and try the other route. Make both targets below.

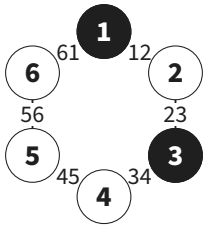


First route: _____

Press count: _____

Other route: _____

Press count: _____



First route: _____

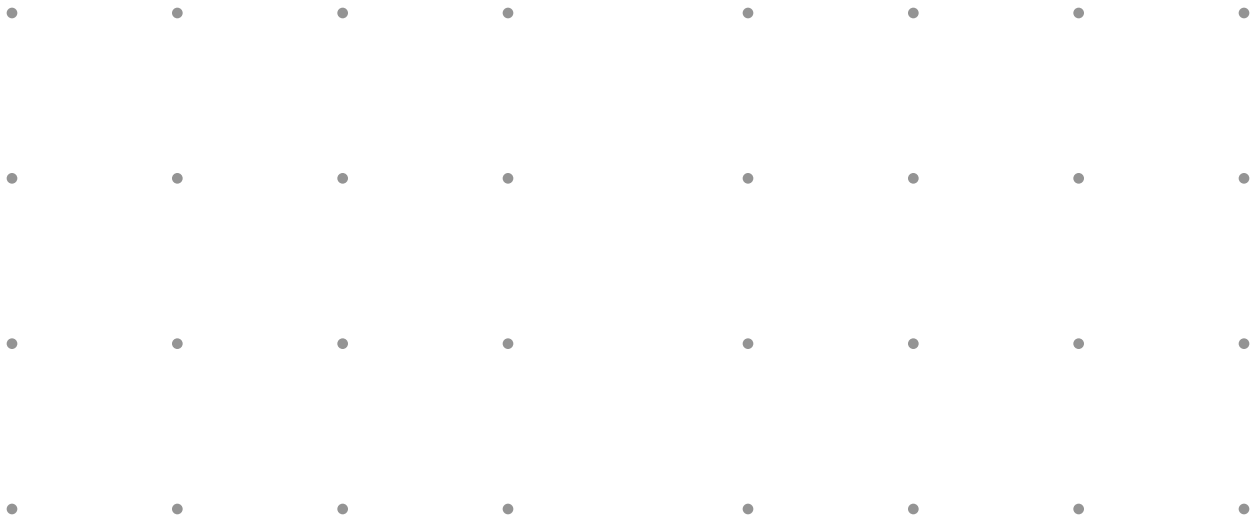
Press count: _____

Other route: _____

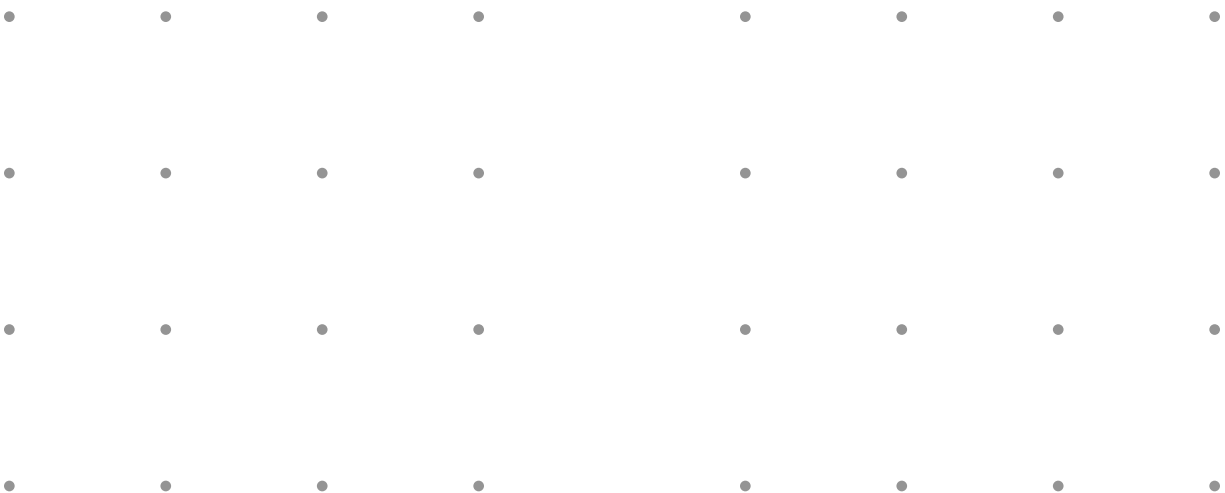
Press count: _____

Circle each pair of routes that uses the same number of presses. Could two routes around the five-lamp ring use the same number? Draw or write your idea.

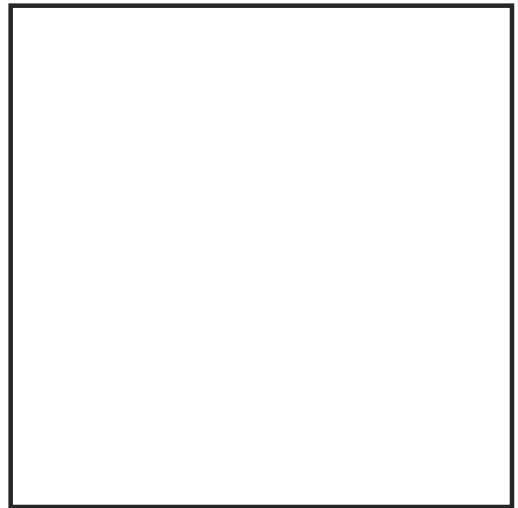
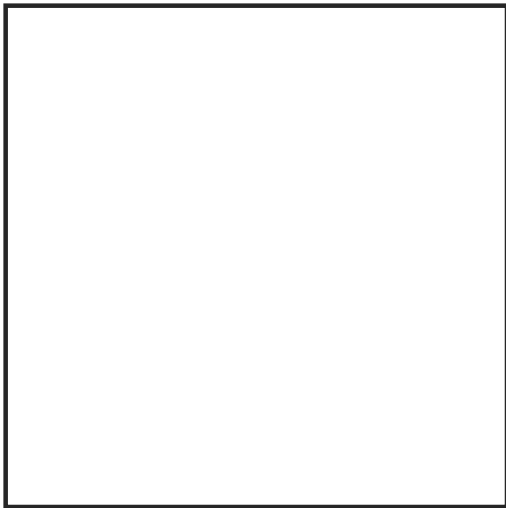
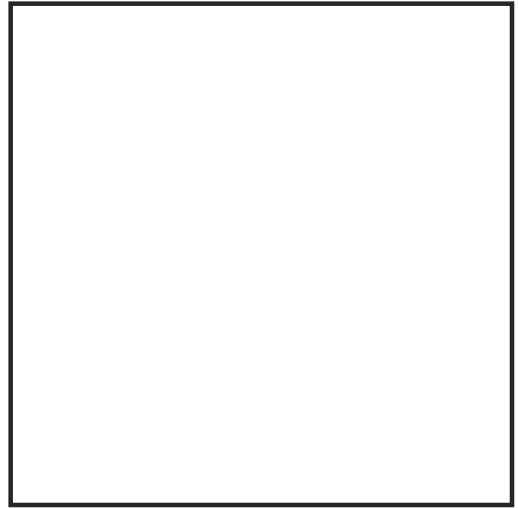
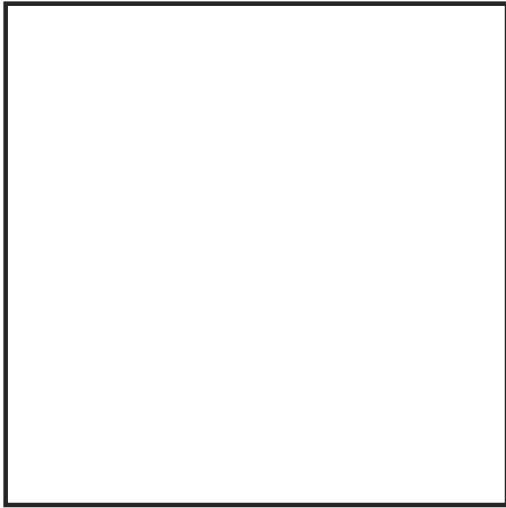
Problem 33: Join dots with straight lines to make a closed shape with THREE corners. Keep your lines from crossing. Make two different shapes.



Problem 34: Now make two different closed shapes with FOUR corners. Touch each corner where your line turns.

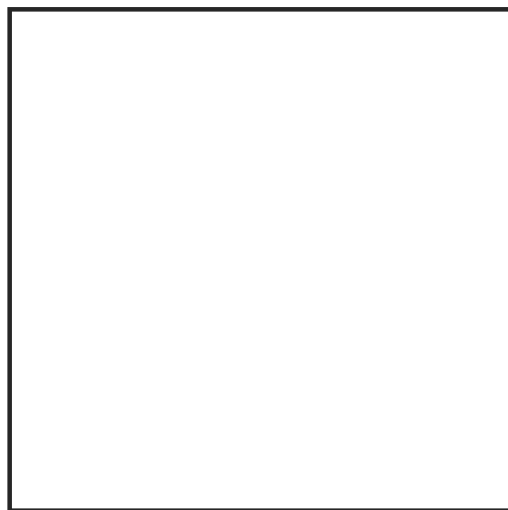
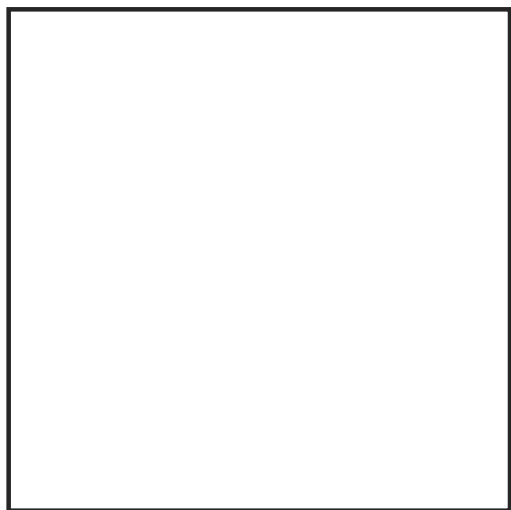
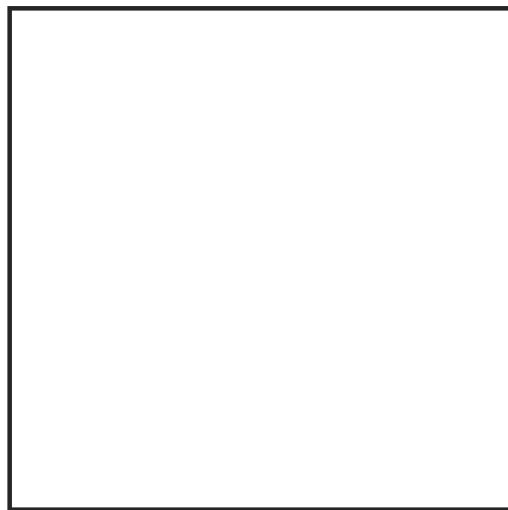
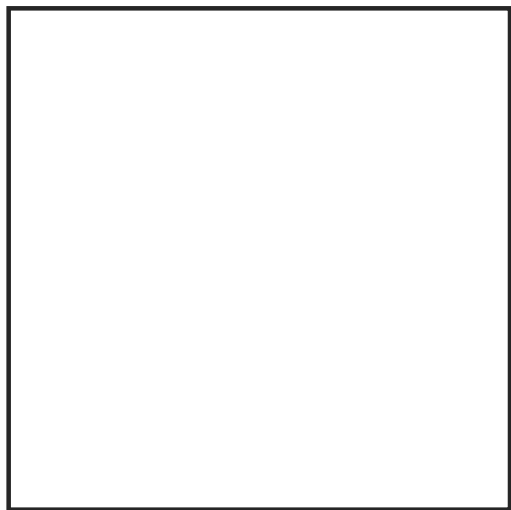


Problem 35: Each square is a house. Draw ONE straight wall all the way across each house, from its outside edge to its outside edge. Make two rooms. Try different walls.

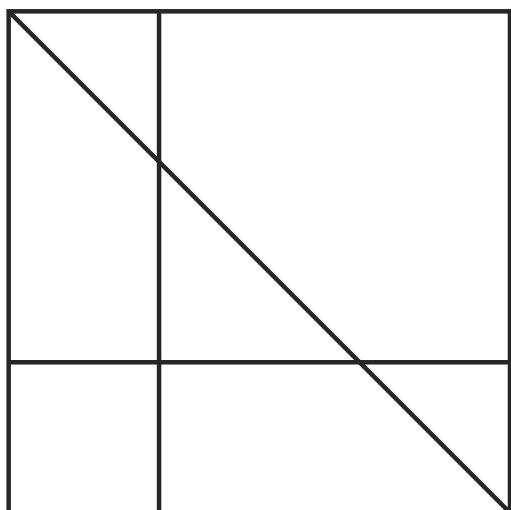
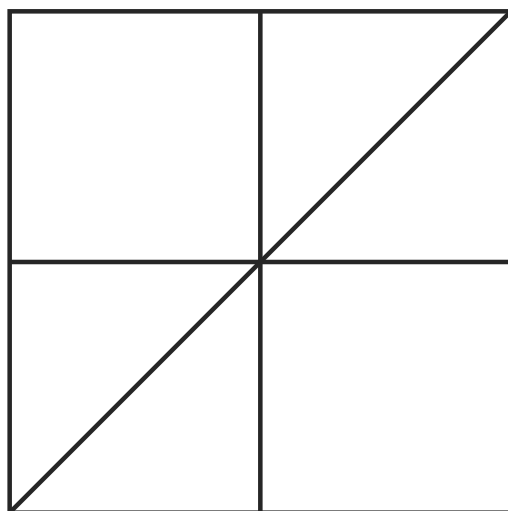
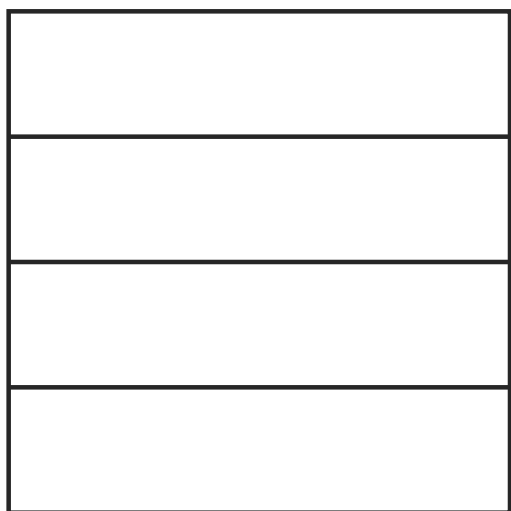


Problem 36: Make or find a house with two triangle rooms. Make or find one with two four-corner rooms. Draw on another sheet if you want to try more houses.

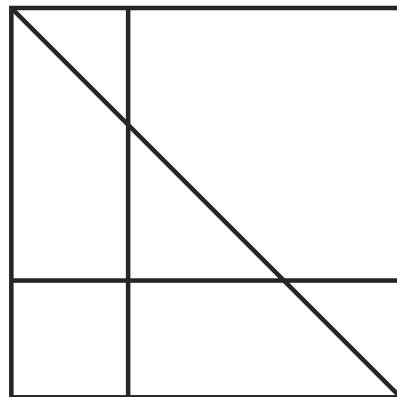
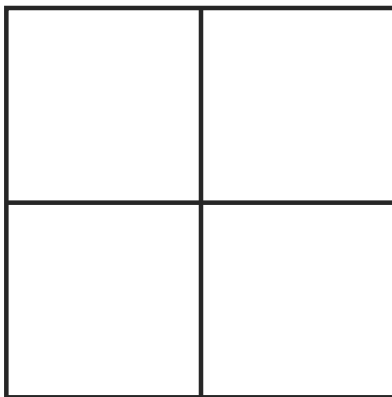
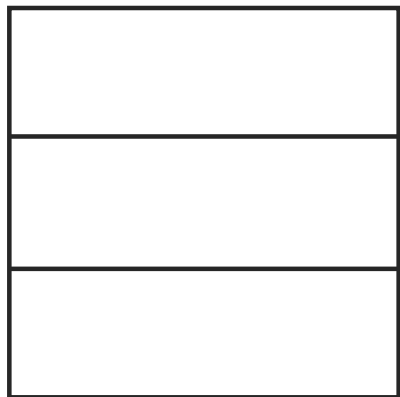
Problem 37: Draw TWO different straight walls all the way across each house. Walls may cross. Put one dot in each room. Try to make different numbers of rooms.



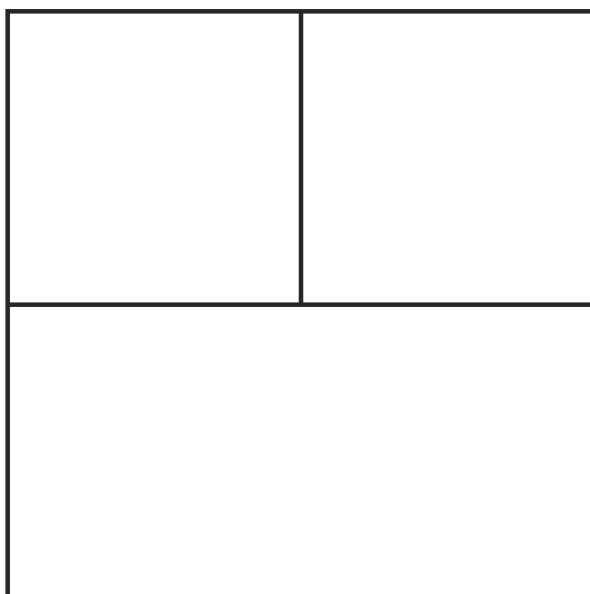
Problem 38: These houses each have THREE straight walls. Put one dot in every room. Which has the most rooms? In the empty house, try three different walls of your own.



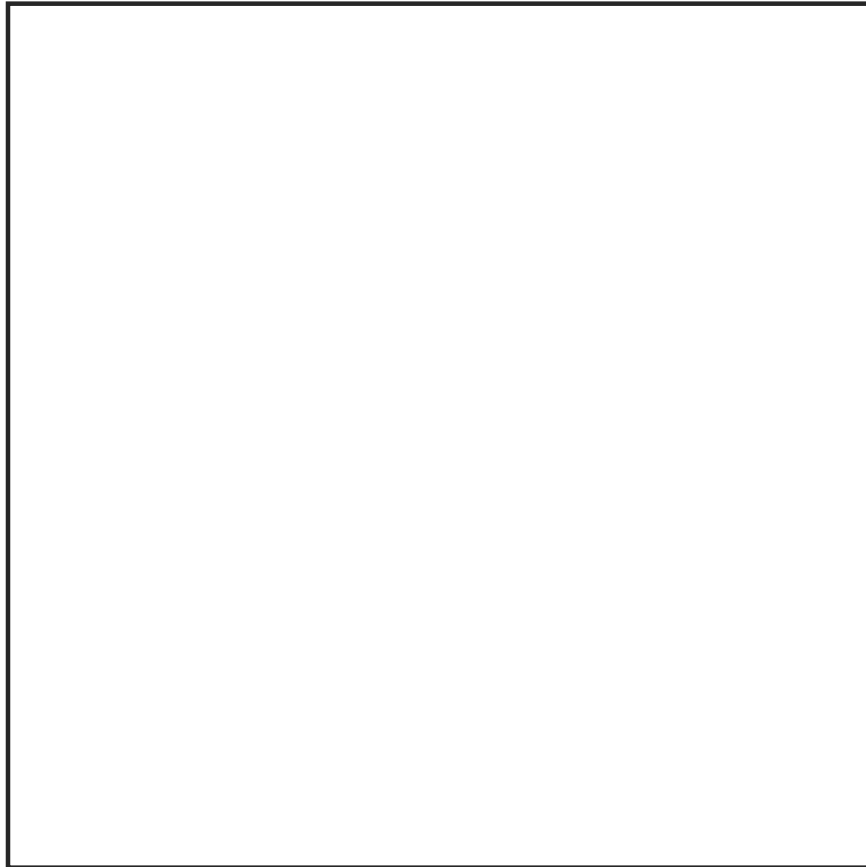
Problem 39: Put a dot or an X in every room in the first three houses. Rooms sharing a wall must have DIFFERENT marks. Rooms touching only at a corner may match.



Problem 40: Try this house. If two marks do not work, you may also use an O.

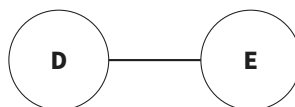
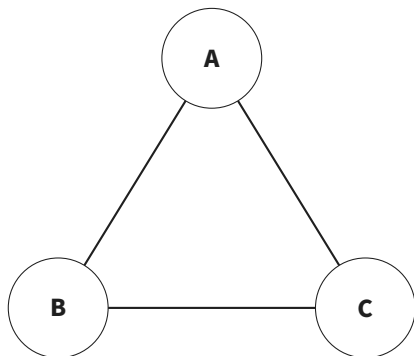


Problem 41: Make a house with one, two, or three different straight walls. Draw each wall all the way across. Let a partner put one dot in every room. Swap jobs.

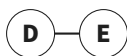
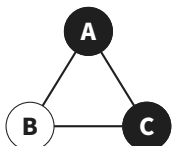


Problem 42: Make a giant house together on a whiteboard or another sheet. Take turns adding one new straight wall all the way across. Do not trace an old wall. Stop after three walls. Find every room together.

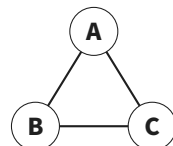
Problem 43: Start all OFF on the large mat. Press a line to flip the lamps at both ends. Make each target below: dark means ON; white means OFF. Use dots you can erase or counters you can flip.



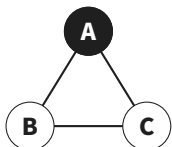
Write **AC** for a press of the line joining A and C. A move cannot jump across the gap. Reset to all OFF for every target.



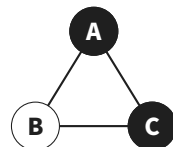
Moves, or “not yet”:



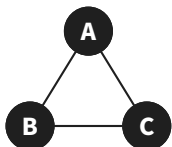
Moves, or “not yet”:



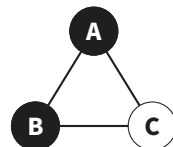
Moves, or “not yet”:



Moves, or “not yet”:

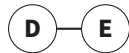
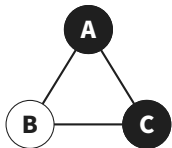
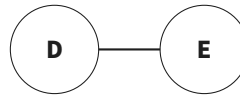
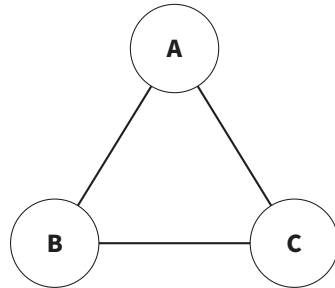


Moves, or “not yet”:

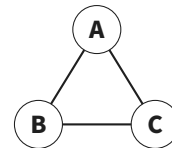


Moves, or “not yet”:

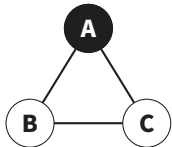
Problem 44: Try these targets, starting all OFF each time. In each picture, count the dark lamps in the triangle. Then count the dark lamps in the pair. Write the two counts and circle “made” or “not yet.”



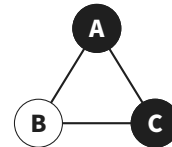
ON in triangle: ____ in pair: ____
Made / not yet



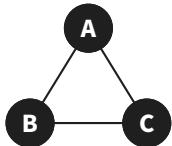
ON in triangle: ____ in pair: ____
Made / not yet



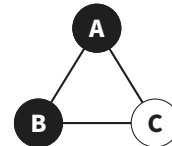
ON in triangle: ____ in pair: ____
Made / not yet



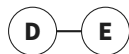
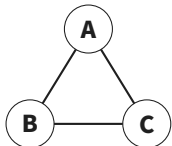
ON in triangle: ____ in pair: ____
Made / not yet



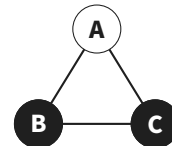
ON in triangle: ____ in pair: ____
Made / not yet



ON in triangle: ____ in pair: ____
Made / not yet



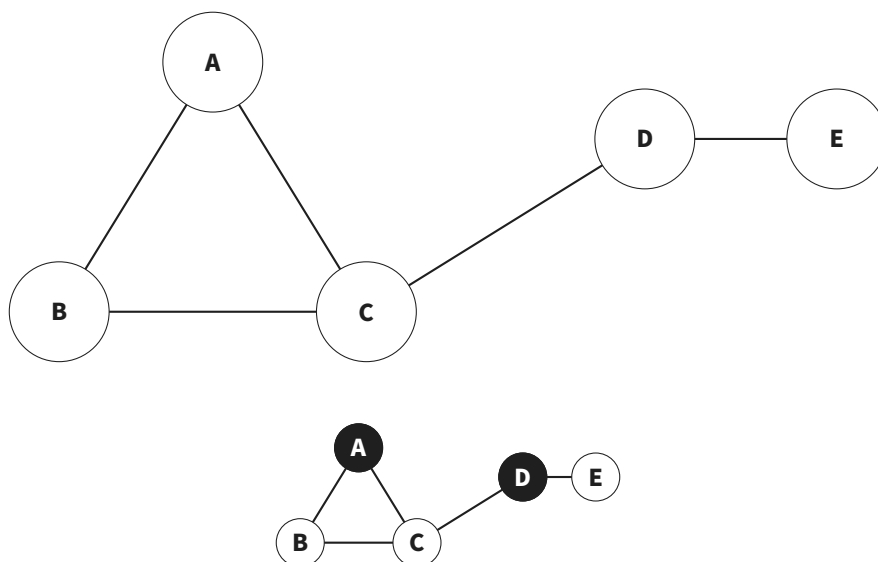
ON in triangle: ____ in pair: ____
Made / not yet



ON in triangle: ____ in pair: ____
Made / not yet

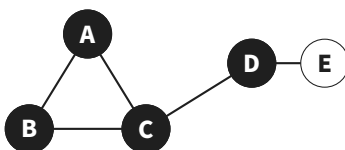
Point to the pictures you made. What do you notice about their two counts? Show a partner a rule that predicts which pictures can be made.

Problem 45: The two pieces now have a bridge: line CD. Start all OFF and try each target below. Press a line to flip both ends. Write the lines you press in order.



Make this picture. Moves: _____

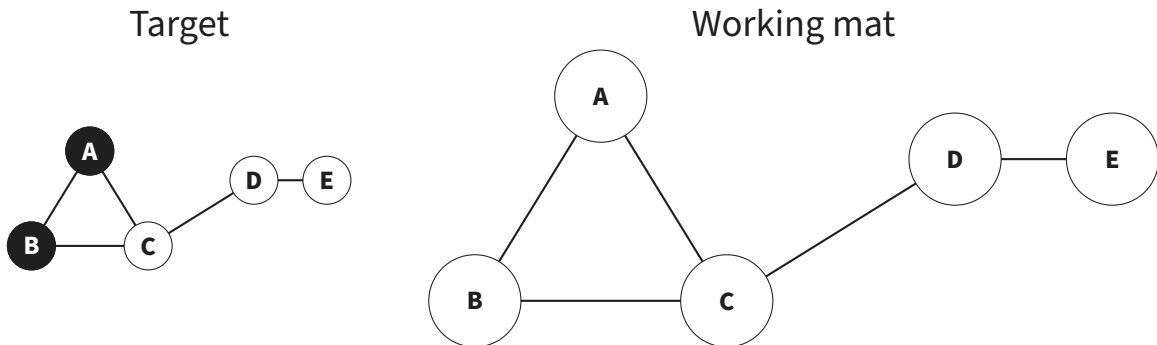
Reset to all OFF. Make the next picture.



Moves: _____

Point to a line your solution needs that was missing from Problem 43. What changed when we added it? Tell a partner, or draw your idea.

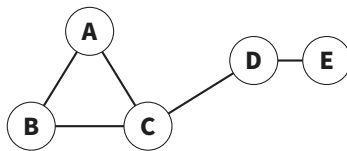
Problem 46: Make the pictured target by trying the two routes shown below. Begin all OFF on the large mat. Press the lines in the first route, then the lines in the second route.



First route: A–C–D. Then: B–C–D.

Try the full list: **AC, CD, BC, CD.**

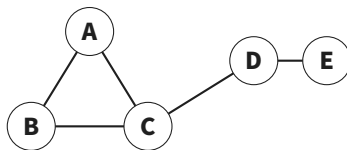
Color the lamps that finish ON in this small picture.



In the list below, cross out both copies of the repeated press.

AC CD BC CD

Reset to all OFF. Try only the presses you did not cross out. Color that final picture below.



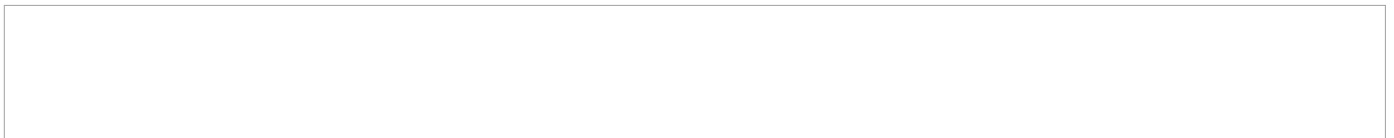
Compare your two pictures with the target. Do both lists work?

Find a longer list of your own that makes this target. Give it to a partner to shorten and test.

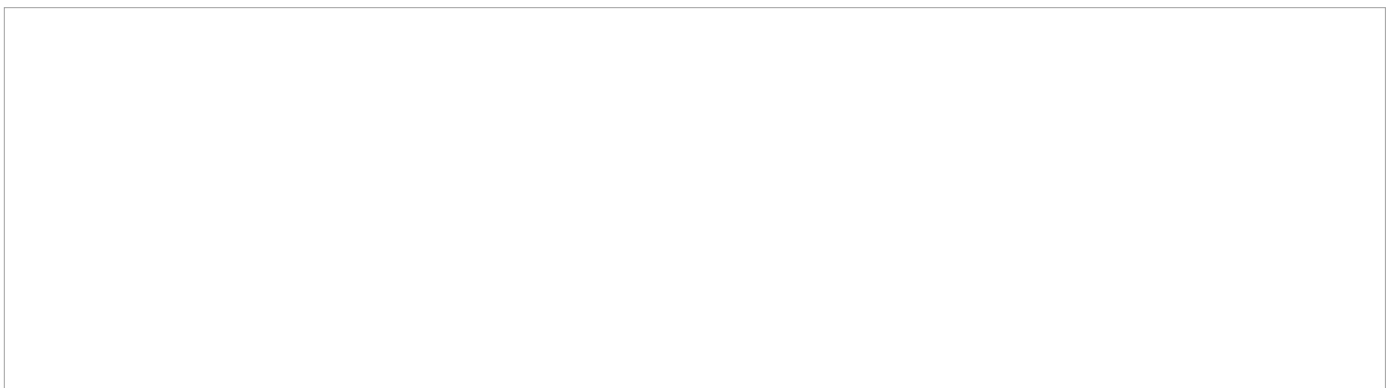
Problem 47: Invent a network with two separate pieces. Draw at least three lamps in each piece, joined by lines. Leave a gap with no line between the pieces.



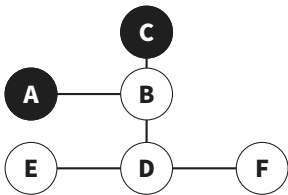
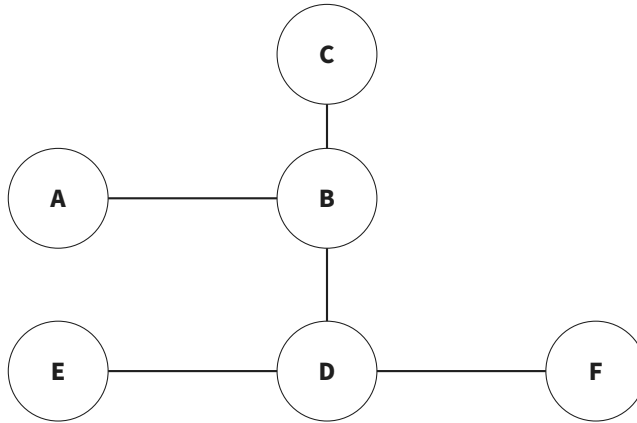
Choose one target you predict can be made from all OFF. Show it by coloring lamps in your drawing. Try it with dots on a whiteboard, or counters on this page. Write a move list or trace the lines you used.



Copy your network below. Color a target you predict cannot be made from all OFF. Ask a partner to try it. Point to the lamps that helped you choose it.



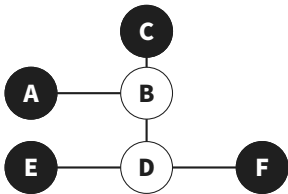
Problem 48: This network has no closed loop. Start all OFF. Press a line to flip both ends. Make each pictured target using every line at most once. Use dots you can erase or counters you can flip.



Make this picture.

Moves: _____

Reset to all OFF before making the next picture.

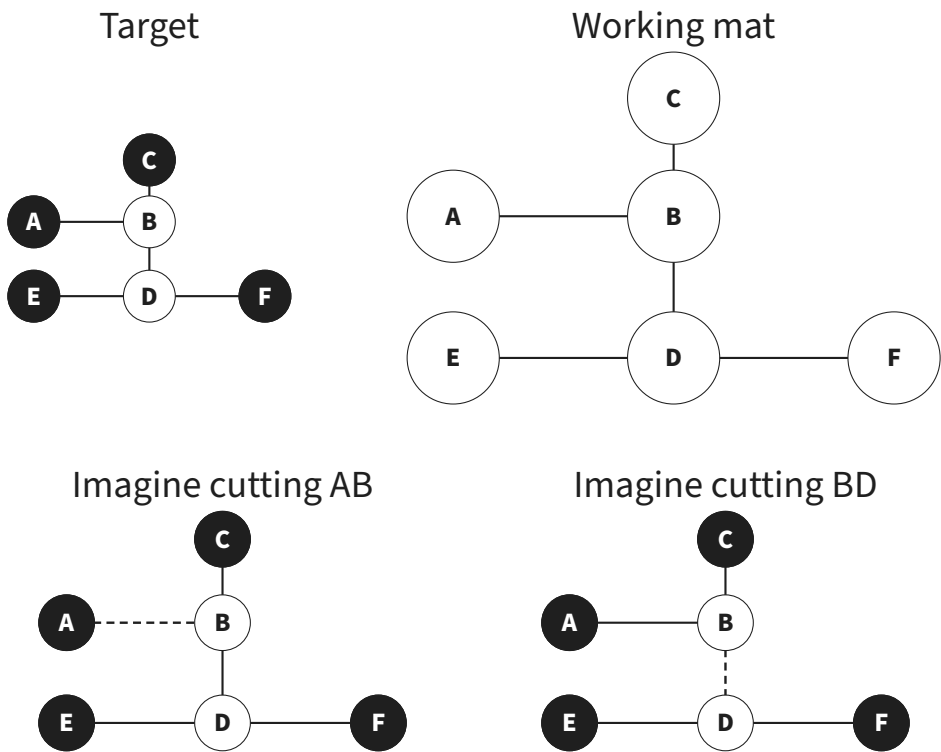


Make this picture.

Moves: _____

Choose one target. Trade move lists with a partner and replay their list. Point to any lines you both used.

Problem 49: Use the target shown below. Imagine cutting one line at a time. Count the dark target lamps on the named side of the cut. Predict which lines to press, then test from all OFF.



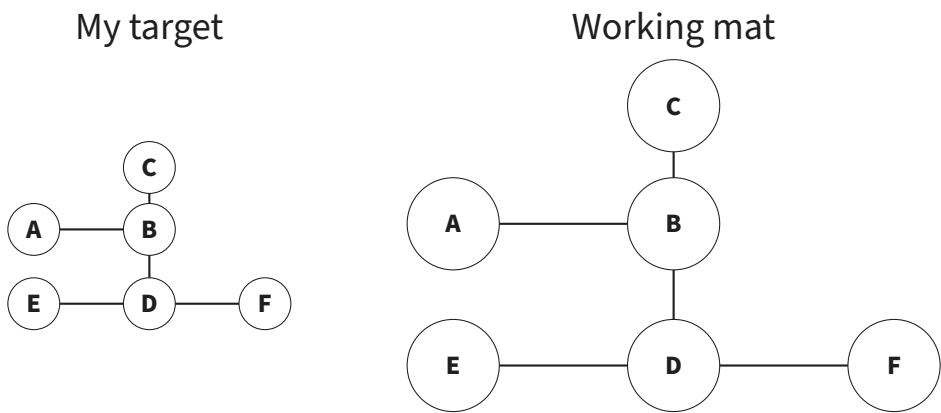
At cut AB, the A side has **1** target lamp. Press AB once. At cut BD, the A,B,C side has **2** target lamps. Leave BD alone.

Complete the other rows. Use each line at most once.

Cut line	Count on this side	Target count	Press?
AB	A	1	yes
BC	C	—	yes / no
BD	A, B, C	2	no
DE	E	—	yes / no
DF	F	—	yes / no

Try your predicted list. Does it make the target? Show a partner how the counts helped you choose the lines.

Problem 50: Invent another target on this tree. Color 2, 4, or 6 lamps ON in the small picture. Choose a different target from Problem 48. Leave the large working mat all OFF.



Before trying moves, imagine cutting each line. Count the ON lamps in your target on the named side. Predict whether to press the line. Use each line at most once.

Cut line	Count on this side	Target count	Press?
AB	A	_____	yes / no
BC	C	_____	yes / no
BD	A, B, C	_____	yes / no
DE	E	_____	yes / no
DF	F	_____	yes / no

Write your predicted move list: _____

Now test it from all OFF. Compare the result with your target. If they differ, circle a row to check again.

Trade targets with a partner. Predict the presses for their target and test them together.

Problem 1: Use cards A–H to make a reversible key: each letter has one output, and each output is used once. Build the four sets of loops below. Move a counter one arrow per turn on each loop. Record the first turn when all counters are home.

Loop sizes	My key or arrow loops	Return
8		
4 + 4		
6 + 2		
5 + 3		

Problem 2: For the 5+3 key, list the return turns of each separate loop through turn 20. Circle the first turn on both lists. Repeat for 4+4. Compare with one loop of size 8.

Problem 3: Find the slowest eight-letter key. Sort possible loop sizes by the largest loop. Use eight cards to check that your sizes total eight. For largest loops 4–8, list the splits of the leftover cards and find the best return. In the last row, use loop sizes 1,2,3 to find one turn that returns every loop. A fixed letter is a loop of size 1.

Largest loop	Possible leftover loops	Best return
8	none	
7		
6		
5		
4		
At most 3		

Problem 4: Compare the best return in every row with your page-1 key. Record the winning loop sizes, key, and return time.

Problem 5: Add card I. Build keys with loops $5+4$, $7+2$, and $4+3+2$. List each loop's return turns until they first agree. Record the whole-key return for each.

Problem 6: Classify nine-letter keys by largest loop, as on page 2. Find the best return for largest loops 9,8,7,6,5,4. When all loops have size at most 3, give one turn that returns all of them. Compare these bounds with your constructions.

Problem 7: Use six cards. Test loops $3+2+1$, 6, and $5+1$. Compare with the five-card $3+2$ key. Predict whether adding a card must always make the longest possible return strictly larger. Record a supporting example and a question still to check.

Problem 1: Lay out cards 0–7, leftmost first. Cut into two equal rows without reversing either half. Interleave one card at a time, starting with the first half. The first shuffle is shown. Rebuild it with cards, then repeat the same rule until the original row first returns. Record every row.

Shuffle	Card labels, left to right
0	0 1 2 3 4 5 6 7
1	0 4 1 5 2 6 3 7
2	
3	
4	

Problem 2: Number places 0–7 from left to right on a strip of paper. Put the original row above it. Track card 1 through the shuffles, recording its place each time. Reset and track card 3, then card 7.

Problem 3: Reset to 0–7. Shuffle once. Fill where each old position moves. A card’s label stays the same; its position can change. For example, the card at old position 4 moves to position 1.

Old position	0	1	2	3	4	5	6	7
New position					1			

Problem 4: Draw arrow loops for this position map, starting at 0 and then at each unused position. Use the loops to predict the first return of the whole row. Check against page 1.

Problem 5: For positions 0–6, double the old position and subtract 7 if needed to leave a number from 0 to 6. Compare with your measured new positions. Treat position 7 separately.

Old position	0	1	2	3	4	5	6
Double							
Remainder after 7							

Problem 6: Starting with 1, repeatedly double and keep the remainder after division by 7. Record until 1 returns. Compare with the route of old position 1.

Problem 7: Use six cards 0–5. Repeat the same out-shuffle until the full row returns. Record rows. Then test whether the position rule is doubling with remainder after division by 5, with position 5 fixed.

Problem 8: For sixteen cards 0–15, predict the return by starting with 1, repeatedly doubling, and taking the remainder after division by 15. Check with cards or draw the position loops; keep position 15 fixed separately.

Problem 9: Compare the first return times for 6,8,16 cards. Record one comparison showing whether a larger deck must take more shuffles to return.

Problem 1: Copy this city. Every row and column must contain 1–4 once. In the upper-left bold block, exchange the 1s and 2s. Record the changed city on the right, then check each row and column.

1	2	3	4
2	1	4	3
3	4	1	2
4	3	2	1

Problem 2: Start from the original city each time. Switch the two heights within just one other bold block. Record each changed city below. Check every row and column after each switch.

Upper right

Lower left

Lower right

Rows and columns affected by my lower-left switch:

Problem 3: An entry clue fixes the height in one cell. There are no outside clues on this page. Copy the original city from Problem 1. Circle three cells as clues. Choose a bold block with no circled cell. Switch its two heights, then check that all three clues still hold. Repeat using a different choice of three clues.

Problem 4: Now keep the four entries shown below. They touch all four bold blocks. Build two different cities that keep these entries and use 1–4 once in every row and column.

1		3	
3		1	

1		3	
3		1	

Cells I changed while keeping the four clues:

Problem 1: Use five counters in a row. Write R as 0 and B as 1. A score counts matching positions. Build each secret below and score all four tests. Keep the secret visible while checking.

Secret	00000	00011	00101	01001
00000				
11111				
10010				

Problem 2: Choose a new secret and hide it. Your partner uses exactly the same four tests. Record the scores. Reveal the secret and check each score. Then swap jobs.

Round 1		Round 2	
Test	Score	Test	Score
00000		00000	
00011		00011	
00101		00101	
01001		01001	

Revealed secrets:

Problem 3: Change two test positions from 00 to 11 while leaving all other positions fixed. Build each possible secret pair below. Count its matches with 00 and with 11, and record the change.

Secret pair	Matches with 00	Matches with 11	Score change
00			
01			
10			
11			

Problem 4: For one hidden five-place secret, tests 00000, 00011, 00101, and 01001 scored 2, 2, 2, and 0. Use the first score to count its 1s. Compare each later score with the first to find the number of 1s in each changed pair.

Positions	Number of 1s
All five positions	
Positions 4 and 5	
Positions 3 and 5	
Positions 2 and 5	

Problem 5: Use those counts to build the secret. Record it and check all four scores directly.

My secret and its four scores:

Problem 6: Try to build a secret with exactly one 1 in each of these pairs: positions 4 and 5, positions 3 and 5, positions 2 and 5. First put 0 in position 5, then put 1 there. In each case try both choices for position 1. Record all four words and their total numbers of 1s.

Position 5	Position 1	Five-place word	Total 1s
0	0		
0	1		
1	0		
1	1		

Problem 7: Score your four words using the four tests from Problem 1. Compare the score records. Circle a score that separates each pair of words.

Secret	00000	00011	00101	01001

Problem 8: Return to your two hidden secrets from Problem 2. Use the total number of 1s and the three pair counts to recover them from the scores alone. Record the recovered words.

Recovered words:

Problem 1: Use two piles. On a turn, take 1, 3, or 4 counters from exactly one pile; do not take more than that pile holds. Players alternate. Taking the very last counter wins. Play each starting pair below. Record the two pile sizes after every move, writing (left pile, right pile).

Start	Pairs after each move	Winner: first or second
(1,1)		
(1,3)		
(1,4)		
(4,4)		

Problem 2: Play again from (1,4). Try every legal first move from either pile. Reset before each attempt. Record which moves leave a position your opponent cannot win against your replies.

First moves and the pairs they leave:

Problem 3: Give each single pile a number label. Set $g(0)=0$. For a larger pile, list the labels of every pile reachable by taking 1, 3, or 4. Its g -label is the smallest nonnegative integer missing from that list. For pile 1, the reachable label is 0, so $g(1)=1$. For pile 2, it is 1, so $g(2)=0$. Build these small cases, then complete the chart in order.

Pile	Legal next piles	Their g -labels	Smallest missing
0	none	none	0
1	0	0	1
2	1	1	0
3			
4			
5			
6			
7			
8			
9			
10			

Problem 4: Choose piles 4 and 5. Build every legal move from each. Use your chart to record the labels reached. Circle any nonnegative label smaller than the starting label that you can reach.

Starting labels and reachable labels:

Problem 5: Use the number labels to revisit two-pile games. For each pair below, write both labels. If they differ, try to make them equal in one legal move. Record a move that does this, then test it in a game.

Piles	Two g-labels	Move, if possible	New pair
(1,3)			
(4,6)			
(5,6)			
(3,4)			

Problem 6: Start from (4,6), whose two labels are equal. List every legal first move and the labels of the new pair. For each new pair, find a reply that restores equal labels.

First move	New pair and labels	My reply

Problem 1: Use two piles. On a turn, remove one or more counters from one pile OR remove the same positive number from both. Take the last counter to win. Play each start twice, changing who starts. Record one first move and reply.



Start	First move leaves	Reply leaves
(2,2)		
(1,2)		
(2,3)		
(3,5)		

Problem 2: From (1,2), try every legal first move. There are four. Draw or list them, then find a reply that wins immediately in each case.

Problem 3: Make a map to find starts worth giving your opponent. T means its next player loses against perfect replies; W means that player can force a win. Start with T at (0,0), where play is over. Work in increasing $a+b$. Mark W when you can move to a T. Mark T only when all legal moves go to W positions. Pile order does not matter; sort the new sizes before locating a move in this table.

$a \setminus b$	0	1	2	3	4	5	6	7
0	T							
1	–							
2	–	–						
3	–	–	–					
4	–	–	–	–				
5	–	–	–	–	–			
6	–	–	–	–	–	–		
7	–	–	–	–	–	–	–	

Problem 4: List the T pairs you found. For each of (2,4), (4,6), and (5,7), build the piles and find a legal move to one of your T pairs. Record the moves.

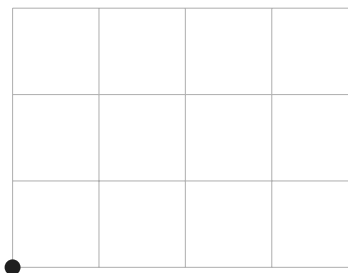
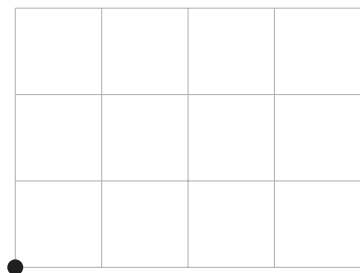
Problem 5: Build a second list. Begin with (0,0). In row n , choose the smallest nonnegative number not yet used in either column as a_n . Set $b_n = a_n + n$. Complete the table, crossing numbers off the strip as you use them. Compare its pairs with your game map.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	
n	0			1				2			3		4		5
a_n	0			1											
b_n	0			2											
$b_n - a_n$	0			1											

Problem 6: Choose two pairs in your list. Try to move from the larger pair to the smaller by taking from one pile, then by taking equally from both. Repeat for two other pairs. Record what prevents each move, or record a move if you find one.

Problem 7: A known theorem gives the losing pairs as $(\lfloor n\phi \rfloor, \lfloor n\phi^2 \rfloor)$, where $\phi = (1 + \sqrt{5})/2 \approx 1.618034$. The brackets mean round down. Use a calculator to check $n=1,2,3,4,5$ against your list. Then find moves into the list from (5,8) and (8,12). Record the moves.

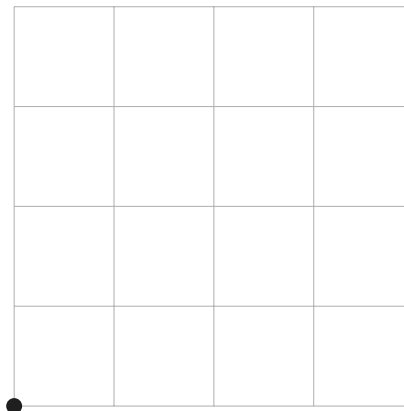
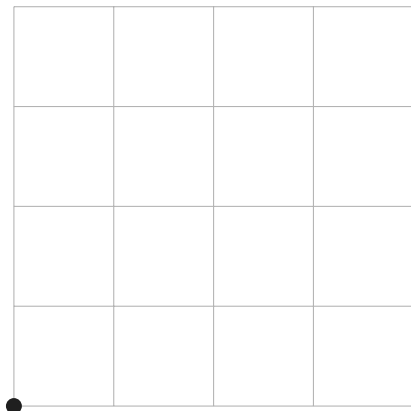
Problem 1: Use a unit square, unfolded into the square grids below. Each little square is one whole reflected table. Start at the black dot. Slope means vertical distance divided by horizontal distance. Draw slopes $1/2$, $2/3$, and $3/2$, one on each grid. Stop at the first grid vertex after the starting dot. Mark each wall crossing before it.



Problem 2: Record each first vertex as (horizontal distance, vertical distance). Fold the whole-room counts back to predict the corner. Count the wall crossings before it.

Slope	First vertex	Corner	Bounces
$1/2$			
$2/3$			
$3/2$			

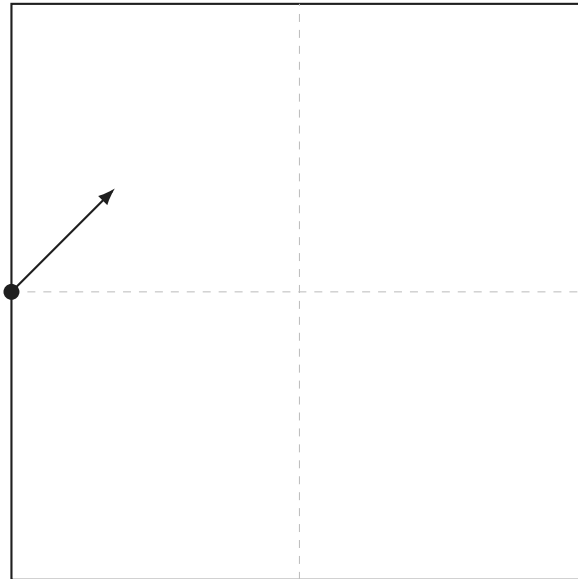
Problem 3: On the left grid draw the line of slope $2/4$. On the right draw the line of slope $3/4$. Stop each at its first grid vertex. Record the vertex, corner, and bounces; compare $2/4$ with your $1/2$ path.



Problem 4: Use positive fractions p/q with no common factor between p and q . Find a slope whose path ends top-right after four bounces. Find one that ends bottom-right after three bounces. Record each slope and its first unfolded vertex, then check the wall counts.

Problem 5: Now use slope $\sqrt{2}$. If its unfolded path from $(0,0)$ passed through a positive integer point (u,v) , its slope would be v/u . Record what this would say about $\sqrt{2}$. You may use the fact that $\sqrt{2}$ is not a ratio of integers.

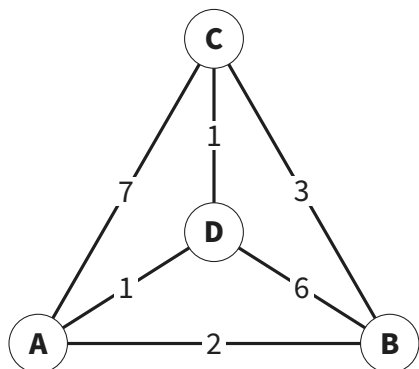
Problem 6: Start at the midpoint of the left wall of this unit square, heading up-right with slope 1. Trace four bounces. Continue for four more. Mark a point inside the square that your path never reaches.



Problem 7: Compare your midpoint path with the slope- $\sqrt{2}$ path from $(0,0)$. For each, record whether it can reach a corner. For the midpoint path, draw a small region that the path never enters.

Problem 8: A known theorem says the slope- $\sqrt{2}$ path from $(0,0)$ comes arbitrarily close to every point in the square. This is called dense. Circle which evidence below establishes density: “I did not hit a corner” / “the stated theorem.” Use your midpoint path to give an example showing why the other claim is insufficient.

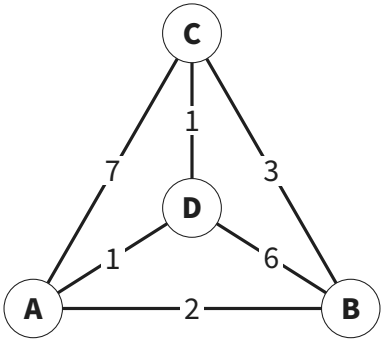
Problem 1: Start at A, travel every road at least once, and return to A. Numbers are road lengths; every repeat costs that length again. Find two complete routes. Record each visited dot and add the traveled lengths. Circle your shorter route.



Trial	Route	Length
1		
2		

Problem 2: Add the lengths of the six original roads. In one of your routes, mark the first trip along each road. List only the extra trips and their total cost. Even a long road must be traveled once.

Problem 3: Find the cheapest connection between each pair of dots. A connection may follow several roads. Compare the direct road with a route through another dot; record the cheapest route and its cost.

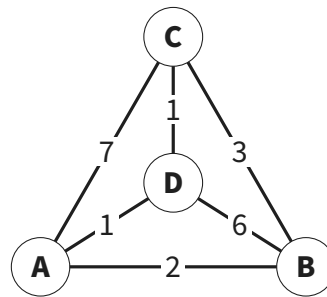
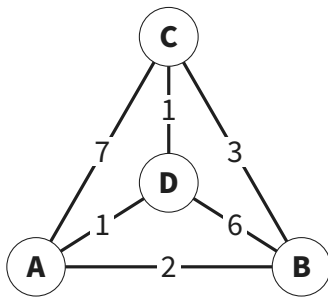


Pair	Direct cost	Cheapest connection	Cost
AB	2		
AC	7		
AD	1		
BC	3		
BD	6		
CD	1		

Problem 4: The four dots all have odd degree. Pair them in each way below and add the two cheapest connection costs. Compare the extra costs. Circle the best pairing.

Pairing	Total extra cost
AB + CD	
AC + BD	
AD + BC	

Problem 5: On the left map mark extra trips for your best pairing. On the right mark extra trips for $AD + BC$. Find a closed route on each map using every original and extra road copy exactly once. Record both routes and total lengths.



Problem 6: A proposed rule is “Choose a cheapest pair first, then connect the remaining two dots.” Choose AD first and calculate the result. Choose CD first and calculate the result. Record whether the rule always finds the best total in this example.

Problem 7: In your best route, erase one required copy of each original road from its crossing list. Draw just the extra road copies below. Circle their odd-degree endpoints and match them with one of the pairings in Problem 4.